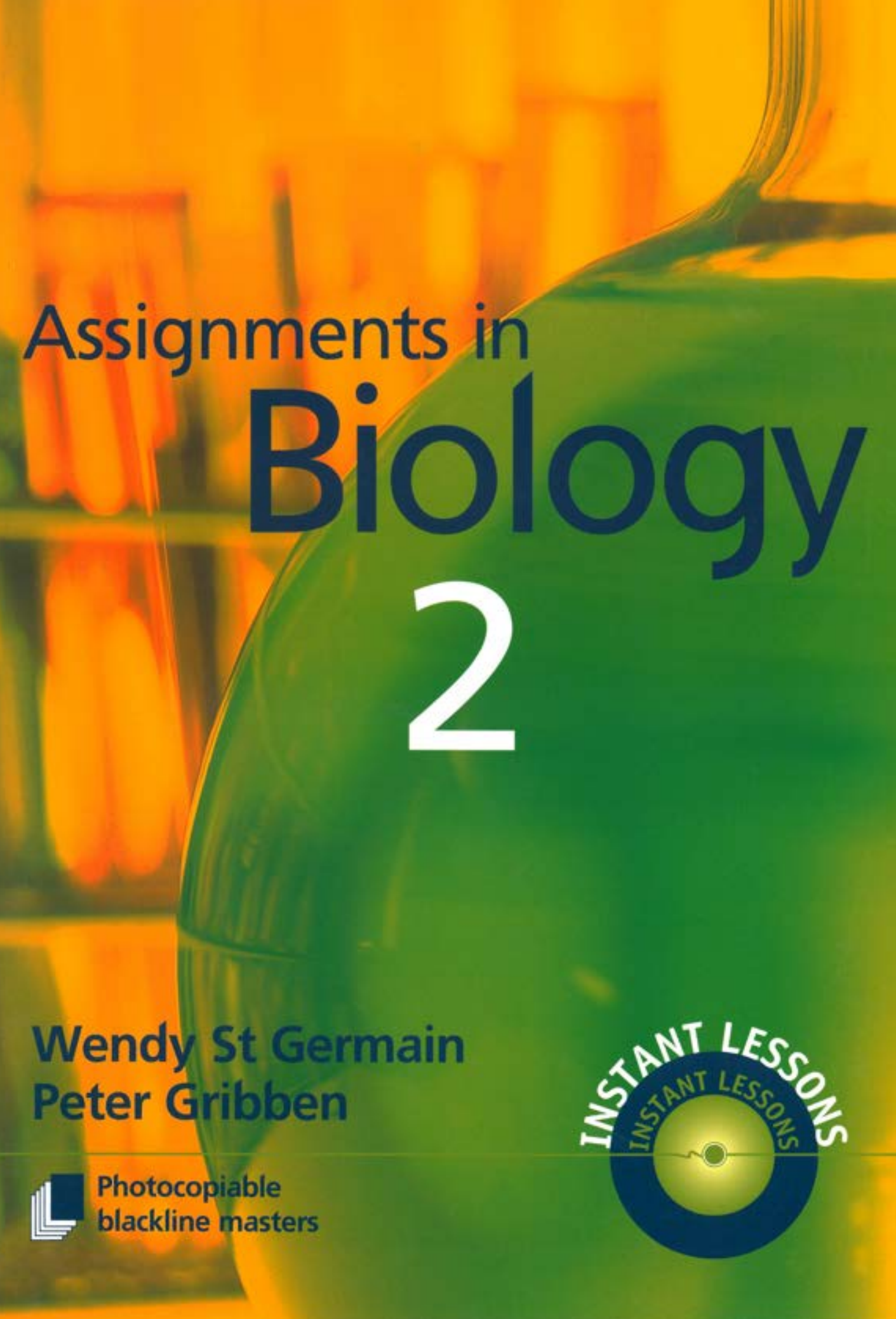




Assignments in **Biology** 2

Wendy St Germain
Peter Gribben



Photocopiable
blackline masters



Assignments in
Biology
2

Peter Gribben
Wendy St Germain



© Peter Gribben, Wendy St Germain and Learning Essentials 2003

Copying of this book for educational purposes

The purchasing educational institution may only photocopy pages within this book in accordance with The Australian Copyright Act 1968 (the Act) and provided the educational institution (or body that administers it) has given a remuneration notice to the Copyright Agency Limited (CAL) under the Act.

It is mandatory, that the ALL photocopies are recorded by the institution for CAL survey purposes.

For details of the CAL license for educational institutions contact:

Copyright Agency Limited
Level 19, 157 Liverpool St
Sydney, NSW, 2000

The Act does not permit the purchasing educational institution to store any part of this book in a retrieval system, or transmit any part of it in any form by any means without the prior written approval of the publisher. All enquiries should be made to the publisher at the address below.

Project Manager: Shelley Barons
Editor: Kristine Bannerman
Design and formatting by Jelly Design
Diagrams by Joseph Lucia
Printed in Australia by Ligare Book Printer

learning
ESSENTIALS
Ready-to-Use Resources

Instant Lessons
An imprint of Learning Essentials
A.B.N. 56 093 980 726
108 Main Rd
Clayton South VIC 3169
Ph: (03) 9509 5366
www.learningessentials.com.au

ISBN 978 1 92561 064 2

Preface

This set of books has been designed to help teachers of biology in most aspects of this subject. Students will find the language, concepts and material much easier to use and assimilate than that in conventional textbooks. The content includes 'stand alone' material as well as sheets that require some existing knowledge and that test these concepts. An understanding of key phrases and concepts is encouraged.

A variety of styles has been used to engage students' attention. Whilst the books have been mainly aimed at senior students, teachers of able year nine and ten students will find much that they can use with these year groups. In order to extend the more motivated students, many topics include a research section. Outline answers are given for these research questions.

The material can be used as reinforcement, revision, homework, and as an introduction to new topics. Replacement teachers who may not be totally familiar with the subject will find these books invaluable.

Note: In most cases space has not been provided for students to answer the questions on the worksheets. Answers should be written in students' usual class work book or folder.



Contents

Reproduction

1 Reproduction in organisms	5
Asexual and sexual reproduction in plants and animals.	
2 Reproduction in plants	7
Male and female parts of angiosperms and their function. Types of pollination.	
3 Human reproduction I	9
The process of fertilisation. Puberty and secondary sexual characteristics. Gestation period of various animals. Multiple births.	
4 Human reproduction II: Birth	11
The developing baby. Childbirth. A cloze passage describing the human reproductive system.	
5 The endocrine system	13
The human body's glands and associated hormones. Representing the menstrual cycle as a timeline.	
6 Methods of contraception	14
Matching different types of contraception with the correct method of action.	
7 Human reproduction review quiz	15
A true or false quiz to revise and consolidate knowledge about human reproduction.	

The Senses

8 Human senses	16
Stimuli, receptors, effectors and responses. Different types of receptors. Visible light. Sense of smell.	
9 Vision I	18
Electromagnetic spectrum. Eye structure. Lenses and correcting vision.	
10 Vision II	20
The role of rods and cones in the retina. Nocturnal and diurnal animals. Colour blindness. Labelling a diagram.	
11 The human ear	22
Structure and function of parts of the human ear.	
12 Sound and communication	24
Frequency and pitch of sound waves. How we make sounds. Vocal communication between animals, fish and insects.	
13 The senses – definitions	26
Matching sensory terms with the correct scientific description.	
14 The senses review quiz	27
A three-part test using multiple choice questions, true and false questions and definitions for revision and consolidation of knowledge.	

Genetics

15 Historic figures in genetics	30
Mendel and his pea experiments. Sutton and Boveri and the chromosome. T.H. Morgan and genes.	
16 Review of genetic terms	32
Consolidation and revision of genetics terms by reading text and providing definitions.	
17 Punnett squares, peas and crossing	33
Review of Punnett's work. Drawing a Punnett square. Defining heterozygous and homozygous.	
18 Role and structure of chromosomes	34
The structure and function of DNA. Genetic manipulation and DNA hybridisation.	
19 Genetic issues in the 21st century	35
Discussion of 'Genethics', gene technology and cytogenetics.	
20 Pedigrees	37
A pedigree table is described in text and diagrams. Interpreting a pedigree. Sex-linkage. The difference between sex and gender.	
21 Genetics – definitions	39
Matching genetic terms with the correction definitions.	

Medical Ethics

22 Medical ethics	40
Comparing Western and Eastern medical ethics. Contemporary ethical issues. Genetic screening.	
23 In vitro fertilisation	42
The steps involved in the IVF process. The first IVF baby. Surrogacy. Pre-implantation Genetic Diagnosis. X-linked diseases.	
24 Blueprint of life crossword	44
Revision of genetics and reproduction in the form of a crossword puzzle.	

Answers	45
----------------	-----------

Reproduction in organisms

1

All living things are made up of *cells*. Cells are the building blocks of life. The simplest forms of life are *organisms* made up of only one cell. All organisms must reproduce in order to make up for losses due to death. If more organisms are being created than are dying at any one time there is pressure for the organism to spread and colonise new *habitats*.

- 1 What is the general term for an organism consisting of one cell?
- 2 Cells are usually described as *microscopic*. What does this mean?
- 3 Give a definition for *habitat*.
- 4 Describe what is likely to happen to organisms that stay in one area which has only limited resources.

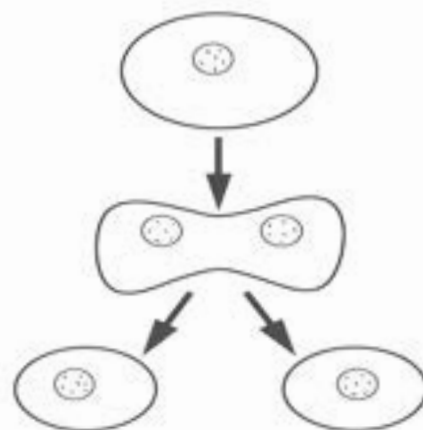
Organisms reproduce by two means: *asexual* reproduction and *sexual* reproduction.

Asexual reproduction

Asexual (meaning 'without sex') is the simplest form of reproduction. There are a number of different types but all share two main features: only one parent is needed and the new individual produced is always genetically identical to this parent.

Some simple creatures (eg amoeba) split into two daughter cells by binary fission. Others (eg yeast and Hydra) form groups of cells that grow out of the organism and develop into a new individual by a process called budding.

- 5 The diagram shows a simplified version of how binary fission occurs. Use the diagram to describe the process.
- 6 A particular *bacterium* (plural *bacteria*) reproduces by binary fission. If this division occurs, on average, once every 20 minutes, calculate how many bacteria theoretically could exist after 12 hours starting from a single bacterium.
- 7 In real life, would you expect the number of bacteria calculated in question 6 to arise? Explain your answer.



Fungi, ferns and mosses produce spores that break off the parent when mature. They are carried about by the wind, by rivers or by sticking to the bodies of animals. Each and every spore is capable of developing into a new individual, provided that it lands in suitable conditions. Even if spores do not find such conditions immediately, they are often capable of prolonged survival until conditions change for the better. Organisms that produce spores create large numbers of them. Since close to 100% of these spores could produce new individuals, this method of reproduction represents great economy of material and effort. However, each spore produced by the organism carries the characteristics of its single parent and will develop into a replica of that parent. There is no variation.

- 8 Identify possible suitable conditions for a spore.
- 9 Name two organisms that can produce spores.
- 10 Give the definition of *offspring*.
- 11 Name two ways that can be used to spread spores.
- 12 Name two types of organism that reproduce by budding.

Plants can also be *propagated*, which involves taking a cutting from one plant and growing that into other (identical) plants.

- 13 What is the name for an organism that is genetically identical to its parent?

Sexual reproduction

Sexual reproduction involves two parent sex cells, which produce sex cells (gametes). For example:

	Female cell	Male cell
Animals	egg (ovum)	sperm
Plants	ovule	pollen

One male cell fertilises one female cell to produce a *zygote*, which is the first cell of a new living organism. The offspring formed by this combination of sex cells possess a mixture of the characteristics of the two parents. For this reason they are always slightly different to both parents. Because of this we get *variation* and variation has been shown to make a species stronger and more able to overcome competitors, survive hard times such as drought, climatic change and other natural disasters.

14 How many parents are involved in asexual reproduction?

15 How many parents are involved in sexual reproduction?

16 Outline why variation of individuals within a species might help that species survive.

Sexual reproduction has two major disadvantages. First, it is slow. The formation of sex cells is a lengthy business and the female cell must wait to be fertilised by male cells. Male cells must find the female cell. Secondly, sexual reproduction is wasteful. All sex cells represent valuable material and effort from the parents. The vast majority of male cells produced fail to find a female cell. They are often spread around at random. For example, the pollen of grass plants is distributed by the wind; 90% of this pollen fails to land on the flowers of a plant of the same species. Because of this many plants and animals have to produce sex cells in large numbers.

17 Identify one advantage and one disadvantage of sexual reproduction.

18 What is a *zygote*?

19 What are *gametes*?

20 Why are organisms formed by sexual reproduction not identical?

21 Why do plants and animals often have to produce so many sex cells?

Organisms in which fertilisation and development of the embryo take place inside the body are called *viviparous* and the baby is born live. In *oviparous* organisms, the development of an embryo occurs inside an egg that has been laid by the mother.

22 Name three viviparous organisms.

23 Name three oviparous organisms.

24 Define *embryo*.

25 Show how the words *viviparous* and *oviparous* are derived. You may need to use a dictionary.

26 Give two disadvantages of asexual reproduction.

27 Give two examples of natural events that might endanger a species.



Reproduction in plants

Many plants reproduce sexually. In plants *pollen* is the male sex cell and *ovum* (plural *ova*) is the female sex cell. Flowers are the organs of sexual reproduction in angiosperms (flowering plants). In sexual reproduction male and female sex cells must come together. This process in plants is called *pollination*.

Flowers are highly modified shoots made up of four major parts. *Sepals*, modified leaves, make up the outer series. The next inner series, the *petals*, are also modified leaves which are usually not green but brightly coloured. The third series consists of the *stamens*, the pollen-producing male portion of the flower. A stamen is made up of an *anther* and a *filament*. The innermost series are the *carpels*, female structures that produce the seeds. The *stigma* is the top of the carpel and is sticky to catch pollen. Carpels can join to form a structure called the *pistil*. The seeds of angiosperms develop in an *ovary*, a part of the carpel that surrounds and protects the egg-containing *ovules*. Seeds develop from the ovules after pollination and fertilisation of the eggs.



Complete the missing parts of the table below. The first one has been done for you.

Flower part	What it does
stamen	male part of the plant, produces pollen
anther	1
2	male reproductive cells
carpel	3
4	inside the carpel, contains ovules
stigma	5
6	develop into seeds after fertilisation
petals	7
8	outer covering or bud

Pollen from one flower is transferred to the stigma of another flower. The pollen can be carried on the bodies of insects or by the wind. Insect-pollinated plants have brightly coloured and scented petals. Wind-pollinated flowers are small, often green and unscented with the anthers and stigmas dangling out of the flower.

9 Give reasons why insect-pollinated plants have brightly coloured and scented flowers.

The pollen joins with an ovum to produce a seed. Seeds contain all the information to grow into a new plant. Seeds need to be spread far away from the parent plant. They can be scattered by the wind, carried in animal fur, eaten and passed out in droppings, some are even dispersed by the fruit exploding. They can also be planted on purpose by humans.

10 Explain why the anthers and stigmas of wind-pollinated plants dangle out of the flower.

11 Give three means by which seeds are moved from the parent plant.



Types of pollination

Pollination is the transfer of pollen from a stamen (male structure of a flower) to a stigma (opening) of the pistil (female structure). The transfer can occur within the same flower (self-pollination) or to a different flower (cross-pollination) of the same species. Different flowers have evolved in different ways to ensure pollination takes place.

12 What are *angiosperms*?

Self-pollination is less dependent on outside forces and more reliable than cross-pollination. However, self-pollination does not create variation between offspring. Some species establish themselves in an area by repeating many copies of the parent plant. If all the offspring are identical, then they run the risk of all being destroyed if their environment changes.

13 Define *environment*.

Seeds that have genetic material from two different plants are more likely to produce more varied offspring that are better able to cope with changes. Some plants have developed ways to make self-pollination less likely. Some plants have 'male' and 'female' versions, with only the male part of flowers on some plants and the female part of flowers on others. In some other types of plants the pollen is produced at a time when the pistil is not ready to receive it.

14 Identify the main male structure of a flower.

15 Identify the main female structure of a flower.

16 Which produces the greater variety of offspring, cross-pollination or self-pollination?

17 Describe two ways in which plants try to stop self-pollination.

Vast amounts of pollen are carried through the air from plants using cross-pollination for fertilisation. Insects, birds and bats are also involved in cross-pollination. They tend to go between flowers of the same species of plant in search of nectar. Bees are especially important in pollination. Often beehives are placed in orchards to ensure that pollination takes place. People are also directly responsible for pollination. Some plant producers hand-pollinate specific plants to eliminate the randomness of wind or insect pollination.

Some people are allergic to air-borne pollen. They sneeze, have inflamed nose and eye membranes and suffer difficulties in breathing.

18 Give a reason why so much pollen is produced by some plants.

19 Name two ways used by plants to attract pollinating creatures.

20 Describe some features of bees that might make them good pollinators.

21 Moths fly at night and find their mates by scent. Give two properties of a plant that will tend to be pollinated by moths.

22 Give a reason why breeders growing plants in glasshouses will often release a bee into the enclosed structure.

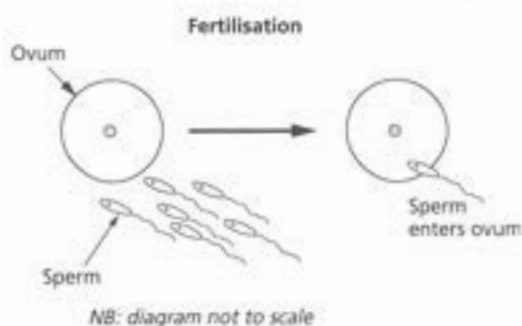
23 Which disease is described by the allergic reaction mentioned above?

24 In Mexico and Central America bees and hummingbirds pollinate vanilla flowers, but in Australia, pollination is done by hand. Give two possible reasons for this.

Human reproduction I

3

The process of reproduction ensures that a species continues. **Fertilisation** is a result of the meeting and joining of a sperm (male sex cell) and an ovum (female sex cell). Semen (produced by males and released during sexual intercourse) contains sperm and is deposited in the vagina, near the cervix. The sperm then 'swim' through the uterus to the oviducts. In an oviduct, one sperm meets and joins the egg. Usually only one sperm fertilises one ovum. Each mature reproductive cell, be it sperm or egg, is called a **gamete**. When the egg has been fertilised, we say **conception** has taken place.



1 What is the plural of ovum?

Sperm are about 0.05 mm in diameter. Ova are about 0.2 mm in diameter. Once released, a sperm cell can live for one to two days and an ovum can live for one day. During this time they can still join together to form a fertilised cell (zygote).

Give definitions for the following terms:

- | | |
|----------|--------------------|
| 2 sperm | 6 gestation period |
| 3 ovum | 7 embryo |
| 4 gamete | 8 foetus |
| 5 uterus | 9 fertilisation |

Puberty

The stage of life when boys and girls first become physically capable of reproducing is called **puberty**. Puberty is the end of childhood and the beginning of adulthood. Puberty occurs at different ages for different people and is controlled by **hormones**. Hormones are produced in organs called **glands**. The ovaries in the female and the testes in the male produce hormones. These chemicals, together with other hormones from the pituitary gland, control sexual development. The pituitary gland is the 'master gland' of the body. Humans must reach a certain weight before puberty can occur.

Usually boys reach puberty between the ages of 13 and 16. At this stage they undergo several physical changes (secondary sexual characteristics).

- Body hair starts to develop.
- The larynx grows so the voice deepens.
- Muscles develop and shoulders become wider.
- Male sex cells start to be produced in the testes.

Usually girls reach puberty between the ages of 10 and 16. At this stage they undergo several physical changes (secondary sexual characteristics).

- Body hair starts to develop.
- Breasts start to develop and hips become more rounded.
- Menstruation (periods) begins, as the cycle of releasing ova begins.

The regular release of eggs means the female body goes through a cycle. This is called the **menstrual cycle**. Menstruation comes from the Latin word *menses*, meaning 'month'. One cycle lasts about 28 days. During the cycle the uterus builds up a lining to nourish a baby. About halfway through the cycle an egg is released from the ovary. The lining of the uterus continues to build up for the next two weeks or so, ready to nourish a fertilised egg. If the egg is not fertilised, the blood-rich lining in the uterus is not needed. The female menstruates, or has a period. The lining and unfertilised egg leave the body. If the egg is fertilised the woman is pregnant and does not menstruate again until after her baby is born. When women reach about 50 years of age they stop menstruating and go through a 'change of life' or **menopause**. After this age eggs are no longer released, certain hormones are no longer present and the woman cannot become pregnant, although rare exceptions do occur.



10 Identify the gland often referred to as the 'master gland'.

Identify the terms described by the following definitions below:

11 *The period when humans become first capable of reproducing sexually.*

12 *A discharge of blood and other material from the uterus of a non-pregnant female at approximately monthly intervals.*

13 *Chemical produced by a gland that affects distant cells.*

14 *The upper part of the trachea that contains the vocal cords.*

15 Outline three secondary sexual characteristics that males gain during puberty.

16 Outline three secondary sexual characteristics that females gain during puberty.

17 At what age do females start menstruating?

Decide whether the following statements are true or false:

18 Girls are born with eggs already in their ovaries.

19 All males go through puberty at the same age.

When an egg is fertilised, it then moves through the oviduct towards the uterus (womb). The egg then embeds itself in the wall of the uterus where it divides and grows. At this stage the developing baby is called an *embryo*. Eight to twelve weeks after fertilisation, the baby looks like a small human. All the main organs of the baby's body are formed. Toes and fingers are starting to form. The baby is then called a *foetus*. The time that the baby is in the uterus is called the *gestation period*. In humans, this is normally about nine months. For a bandicoot it is two weeks, a quoll three weeks, a kangaroo one month, a cat nine weeks, a dog nine weeks, sheep five months and an elephant 22 months.

To survive, the developing baby needs:

- food and oxygen
- protection
- its wastes removed.

20 Describe how these three needs are met while the baby is developing in the womb.

Multiple births

The majority of human births are singletons – ie only one child is born. Sometimes more than one baby can result from a pregnancy, in the case of twins, triplets, quadruplets etc. These multiple births can be due to natural causes or can be a result of assisted reproduction. Inducing ovulation by chemical means increases the chances of a multiple birth by 8–20% and using in vitro fertilisation increases the chances by 20 – 25%. The table below shows the figures for a typical Australian state in one particular year. Study the table and answer the following questions.

No. of confinements	No. of babies born (total)	Singleton births	Sets of twins	Sets of triplets
640	771	515	116	8

21 What percentage of confinements resulted in:

- a singletons?
- b twins?
- c triplets?

Human reproduction II: Birth

An embryo obtains food and oxygen from the mother's blood via the *placenta*, which also removes waste. The placenta forms on the wall of the uterus as the embryo develops. The placenta and embryo are joined by the *umbilical cord*. The blood of the mother and baby do not meet or mix. The two blood supplies are separate.

- 1 Will a baby necessarily have the same blood group as the mother?
Explain your answer.

The baby floats in a bag of liquid called the *amniotic fluid*. The fluid acts as a shock absorber and protects the baby. It cushions the baby from minor physical knocks the mother may receive. Samples of the amniotic fluid containing cells of the unborn baby can be used to check the sex of the child or for early detection of certain disorders.

After about nine months the baby is usually fully developed and in the correct position to be born. Hormones within the mother's body signal the uterus to begin contractions, or *labour*. These contractions start to push the baby out.



The baby passes head-first, usually face-down, through the cervix and down the vaginal canal. After the baby is born, the umbilical cord is tied and cut. The navel, or belly button, is where the cord was attached. There are no nerves in the cord, so it does not hurt the baby when the cord is severed. The placenta (or afterbirth) comes away from the mother soon after the baby is born. It is checked for abnormalities and weighed (it should weigh about the same as the baby, otherwise it may not have been fully expelled). It is then discarded, unless permission has been given to use it for medical purposes, such as the extraction of stem cells.

Some babies are not in the upside-down position when they are ready to be born. These babies are either born bottom first in a breech birth, which can cause injury to both mother and baby, or by caesarean section. A caesarean section involves surgically removing the baby by cutting through the mother's abdominal wall, while the mother is under a general anaesthetic.

- 2 Describe the function of an anaesthetic.
- 3 Give another name for the navel.
- 4 What causes contractions in the uterus to start?

Identify the terms described by the following definitions:

- 5 Cord coming from the navel connecting the foetus with the placenta.
- 6 Organ that gives food and oxygen to foetus and removes carbon dioxide and waste.
- 7 Delivery of a baby by cutting mother's abdomen and uterus.
- 8 Narrow, outer end of uterus.
- 9 The fluid in which the embryo is suspended within the amnion.
- 10 Organ of the female mammal that holds developing embryo.
- 11 Process of giving birth.

Research

- 12 Outline precautions that pregnant women should take to avoid harm to their unborn child.

13 Use words from box to fill in the spaces.

AMNIOTIC	AFTERBIRTH	ANIMAL	BABY	CELLS	CERVIX	CYTOPLASM
DIVIDE	EMBRYO	FERTILISATION	GROWS	IMPLANTATION	LIFE	MICROSCOPE
NAVEL	NUCLEUS	ORGANISM	OVARIES	oviduct	OVUM	OXYGEN
	PLACENTA	SPERM	TESTES	UMBILICAL	UTERUS	ZYGOTE

All living things are made up of _____. Cells are the building blocks of _____. The simplest form of life is a unicellular animal, or _____, with one cell. You need a _____ to see cells properly. Plant cells and _____ cells are different. The _____ controls all the activities of the cell. The cell membrane holds the contents of the cell and the _____ is the material in the cell apart from the nucleus.

Male sex cells are called _____ and female sex cells are called _____ (plural ova).

When a sperm joins with an ovum, _____ takes place forming a new, single cell called a _____. Ova (eggs) are produced by the _____ of a female and pass through the _____. Sperm cells are produced in the _____ (testicles) and are passed out of the _____ during sexual intercourse when the penis is placed inside the vagina. Sperm move through the _____ (or womb) and, if an ovum is present, fertilisation can take place.

The zygote cells then _____ repeatedly and it moves towards the uterus. It becomes embedded on the wall of the uterus – this is called _____. The fertilised cells develop further to become an _____, then a foetus and eventually to be born as a baby. As the embryo _____ in the womb it is nourished and protected. It is surrounded by a sac filled with _____ fluid. During pregnancy an organ called the _____ develops. This allows food and _____ to go to the developing foetus and takes away waste products. The placenta extends into the _____ cord. The umbilical cord is a cord of blood vessels joining the _____ of the foetus to the placenta.

Approximately nine months after fertilisation, the baby is born. The uterus wall contracts, the amniotic fluid, in which the baby has been floating, escapes and the _____ is pushed through the _____ and vagina. The umbilical cord is cut, the placenta is expelled as the _____ and the baby starts to use its own lungs.

The endocrine system

The endocrine system helps control our bodies by secreting (releasing) hormones from glands. Some hormones only act on specific body parts, others cause a more general response. The principal controller of hormone production is the hypothalamus (a part of the brain). The main glands concerned in reproduction are the pituitary, testes and ovaries.

The pituitary is located at the base of the brain and is less than 1 cm in diameter and 0.5 g in mass. As well as releasing hormones that control growth and reproduction it manufactures others, which, in turn, control many other glands.

The gonads are where gametes are produced. The male gonads are the testes and the female gonads are the ovaries. Hormones from the pituitary gland and the steroid hormones from ovary or testis control reproductive function in both sexes. In women, fluctuations of these hormones govern each menstrual cycle, stimulating ovulation and preparing the uterus for pregnancy. The hypothalamus causes the pituitary gland to produce gonadotropins – hormones that stimulate gonads.

The sex steroids also bring about the development of secondary sexual characteristics. In males, androgens cause beard growth, deepening of the voice and increase in muscle size. Oestrogen and progesterone cause breast development and the menstrual cycle in females. The male steroids are anabolic (building up) and are among the banned substances that some athletes have used to encourage muscle development and enhance performance.

1 Give reasons why athletic and sports organisations would ban their members from taking anabolic steroids.

When puberty is reached, humans produce follicle stimulating hormone (FSH) and luteinising hormone (LH). FSH causes the formation of sperm in males and stimulates the development of ova in females. LH causes the production of androgens in males and for females stimulates ovulation. FSH starts off the menstrual cycle.

If an egg is not fertilised the cycle is approximately as follows:

- First 5 days: menstruation, the lining of the womb breaks down and is shed.
- Next 7 days: an egg matures in the follicle.
- Next 3 days: ovulation – egg is released into fallopian tube.
- Next 7 days: egg dies if not fertilised.
- Next 6 days: uterus lining continues to thicken before it starts to break down.

Complete the following table.

Gland	Location	In male or female	Produces
testis	groin	2	3
4	base of brain	5	range of hormones
ovary	6	7	8

9 What is a gonad?

10 Represent the menstrual cycle on the timeline below.

The menstrual cycle

Methods of contraception

Contraception is the deliberate prevention of conception or impregnation, ie stopping a female from becoming pregnant. There are many means of contraception. The boxes below contain the names of methods of contraception and details of how they work. Cut out the boxes and divide them into two groups: 'Type of contraception' and 'Method of action'. Match up each type of contraception with its method of action.

1 Cap or diaphragm	10 Rhythm method
2 Observing changes in vaginal mucus to determine time of ova release.	11 Plastic or metal item placed in uterus, which prevents implantation of zygote.
3 Vasectomy in males and tubular ligation in females.	12 Fitted in the uterus to prevent sperm from passing into the oviduct.
4 Male withdraws penis from vagina before ejaculation.	13 Worn by male to prevent discharge of semen in vagina.
5 The 'pill'	14 Ovulation
6 Condom ('sheath')	15 Intra uterine device (IUD)
7 Foam or jelly placed in vagina, which destroys sperm.	16 Abstaining from sexual intercourse around the time of ovulation.
8 Coitus interruptus ('withdrawal')	17 Prevents ovulation by keeping a high level of oestrogen and progesterone.
9 Sterilisation	18 Spermicide

Human reproduction review quiz

7

Read the statements carefully. Decide whether you think they are *true* (T), *false* (F) or if you are *unsure* (U).

After a class discussion, write interesting or important comments that arose under the relevant statements.



- 1 Gametes are cells involved in reproduction.
- 2 A sperm is a male sex cell.
- 3 A female sex cell is an ovum.
- 4 Another name for the uterus is 'womb'.
- 5 A zygote is a fertilised egg.
- 6 All ova are fertilised by sperm.
- 7 During pregnancy females develop an organ called the placenta.
- 8 The umbilical cord connects a foetus to the placenta.
- 9 Usually boys achieve puberty between the ages of 18 and 21.
- 10 Hormones are produced only by organs in males.
- 11 Testes are glands not organs.
- 12 Everyone reaches puberty at the same age.
- 13 Sex cells are too small to be seen even with a microscope.
- 14 A human embryo develops into a foetus and then is born as a baby.
- 15 A sperm is much larger than an ovum.
- 16 The gestation period is the time it takes a sperm to fertilise an ovum.
- 17 The developing baby floats in a bag of liquid called amniotic fluid.
- 18 All mammals have the same gestation period as humans.
- 19 Another name for the afterbirth is the placenta.
- 20 Contraception is a means of increasing fertility in males and females.

Human senses

If a person is to survive and function properly, he or she must respond to changes in the environment. Anything that causes a response from us is called a *stimulus* (plural *stimuli*). We detect stimuli from both inside and outside our bodies and act upon them. The parts of the body that detect these stimuli are called *receptors*.

1 Name three types of stimuli that a person might experience in normal living.

Receptors are groups of specialised cells that respond to stimuli. Sense organs, eg eyes, ears, tongue, nose and skin, contain many receptors. A stimulus causes these cells to send information to other parts of the body using a *messenger*, either through the bloodstream or by electrical pulses in the nervous system. When the signal reaches an *effector* at a specific location, there is a response. The effector, usually a muscle or a gland, does something.

2 Describe how a muscle might respond when it receives a signal.

3 Describe how a gland might respond when it receives a signal.

The sequence is: stimulus → receptor → messenger → effector → response.

The table below shows some stimuli, how they are detected and the sensation or response they cause.

Stimulus	Location of receptor	Sensation/response
light	eye	sight
sound	ear	hearing
warmth	skin	warmth
cold	skin	cold
touch	skin	pressure/pain
chemicals	mouth, nose	taste, smell

Copy and complete the table. The first one is done for you.

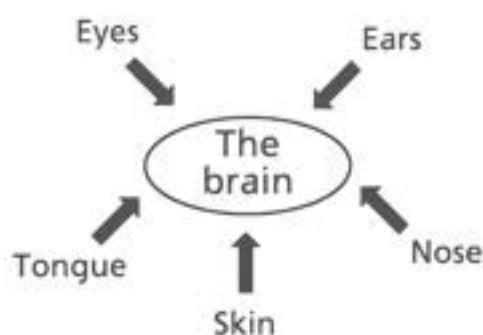
Stimulus	Sense organ	Response
speech	ear	turn towards speaker
bright light	4	blink
heat	5	sweat
6	mouth	taste
chemicals	7	smell
8	skin	pressure
gravity	9	balance

10 Identify the definition that best describes a **stimulus**.

- A An incentive for a person to do something.
- B Something that excites a sensory organ, causes a muscle to contract or a gland to secrete.
- C A chemical messenger sent by a gland through the bloodstream.
- D Part of the body that registers changes in the immediate environment.

Research

- 11 Describe what *sensory deprivation* is and how it affects humans.
- 12 The colours of our surroundings are supposed to influence our moods. Outline some theories about this and give some examples.



The brain is being bombarded by messages from the various sensory systems all of the time. It has to sort out which messages it can ignore from those which are important. Then it has to decide what to do about the important ones. It could store the information or it could choose between several different ways of reacting. It is this ability to think and make complicated choices that makes our brain different from that of other animals.

Sensory receptors are made up of several groups. Copy the following table and state what the receptors sense and where they are located.

Receptor type	What receptor responds to	Location of receptor
chemoreceptor	13	14
hair cells	15	16
mechanoreceptor	17	18
photoreceptor	19	20
thermoreceptor	21	22

We use our senses to take in information from our external environment. One way we do this is by communicating with each other. Sound is used when we talk to each other.

23 Give a definition for *communicate*.

24 Give two other examples where (non-speech) sound is used by humans to communicate.

It is likely that when you describe your surroundings, the sense that you use the most is *sight*. It has been estimated that in humans, of the information that the brain uses, 70% comes from sight, 10% from smell, 5% from taste and 5% from touch. Sight is used in communication when we read or look at pictures or gestures etc.

Visible light is sometimes referred to as 'white light'. It is the part of the electromagnetic spectrum that the human eye uses to see things. The eye collects visible light, changes this to electrical signals and these are carried by nerves to the brain.

In humans, the sense of *smell* is relatively underdeveloped when compared to other animals, for example a dog or a pig, which depend on it to a far greater extent. Substances that have an odour give off molecules, which are present in air and are breathed in. Mucous membrane, a damp lining in the nose, can dissolve these small particles. The special nerve endings present pass the message to the brain.

State whether the following statements are true or false:

25 'Aural' relates to the ear or to the sense of hearing.

26 Our ears contain balance receptors.

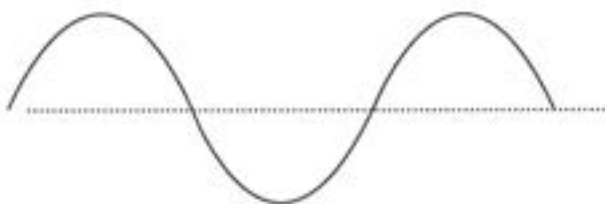
27 The most important human sense is touch.

28 You have been watching a programme on TV about 'ESP – Extra Sensory Perception'. In this show some people claimed they could read other people's minds. You would like to test to see how likely this is. Describe an experiment you could carry out, using, for example, shapes on a series of cards.

Vision I

Visible light is part of the electromagnetic spectrum and is made up of waves of different wavelength. Electromagnetic waves carry energy and travel in straight lines at 300 000 kilometres per second (km s^{-1}) in a vacuum. (A vacuum is an enclosed space with a total absence of any matter.) As the name implies, electromagnetic waves are electric and magnetic ripples caused by the vibration or movement of particles. In order of increasing energy, the electromagnetic spectrum includes: radio waves, microwaves, infra-red, visible light, ultra-violet, X-rays, gamma rays and cosmic rays. Their wavelengths start at 10^8 metres, to smaller than 10^{-11} metres. Wavelength is the distance from peak-to-peak or trough-to-trough in a wave.

- 1 Copy the diagram of a wave, and indicate its wavelength on the diagram.



Visible light has wavelengths from around 400 nanometres ($\text{nm} = 10^{-9} \text{ m}$) at the violet end of the spectrum to around 700 nm at the red end of the spectrum. Enormous numbers of animals have evolved specialised structures that are sensitive to visible light. The light receptors in many invertebrates don't form images in the same way that the human eye does, but just respond to the presence and intensity of light.

In vertebrates, eyes are the organs of sight, sending nervous impulses to the brain when stimulated by light rays. The brain interprets these impulses to produce

images. Each eye consists of a hollow sphere (eyeball). It is set into a socket in the skull and is protected by eyelids and eyelashes. Muscles join the eyeball to the eye socket. They contract to enable the eye to swivel around. Glands at the top of each eye release a watery fluid that keeps the surface of the eyes moist and clean.

- 2 What do we call this 'watery fluid' that bathes the eyes?

Animals with complex eyes that can form images often see the world in a different fashion to humans. Their visual acuity (how sharply they can see things; keenness of perception) will also be different. The standard definition of normal human visual acuity is 20/20 vision. (This originally came from testing vision at a distance of 20 feet, or just over 6 metres.)

Humans, like most animals, have two eyes. Two eyes are enough to give us a good chance of survival in the wild. How the eyes are placed is also important. Animals such as rabbits, horses and birds have eyes on the sides of their head. This enables them to see in an almost complete circle and gives them a good chance to see predators sneaking up on them. However, side eyes are no good for judging distances. If both eyes are at the front of the head, it is called *binocular vision*. Each eye sees a slightly different version of the same object and so the brain can make fairly accurate estimate about how far away the object is, using the angle of the two eyes and by combining the images from each eye.

- 3 Give a definition of *predator* and give 3 examples.

Identify these parts of the human body.

- 4 A pair of spheres each measuring about 2.5 cm and set into sockets in the skull.
- 5 Large organ that coordinates stimuli from sense organs and acts on these stimuli.
- 6 Organ that secretes chemicals for use in the body.
- 7 Outline the differences in the vision of an animal with eyes at the side of its head compared with an animal with eyes at the front.

Research

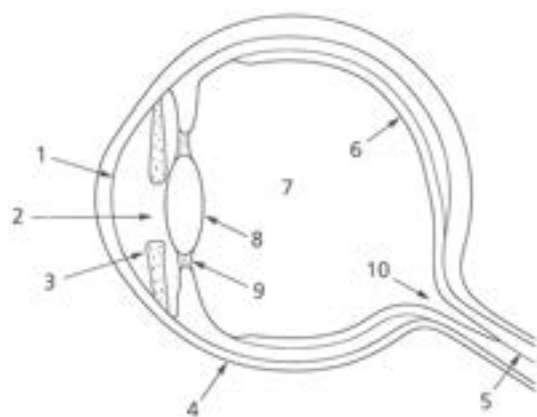
- 8 Give two examples of how human vision differs from a named animal and give a possible reason for this difference.

The eye

The front part of your eye – the clear outer part – is the *cornea*. It is very sensitive and must be kept clean. Whenever your eyelids close, a watery fluid is released by the *lachrymal glands*. Tears contain salts and an enzyme that has an anti-bacterial action. This fluid helps to remove dust and other small particles that could irritate or damage the eye. Your eyebrows and eyelashes also play an important part in protecting your eyes from being damaged by water and small particles. If you get something in your eye, you should not rub it. This moves the particles further into the eye, making it more difficult to remove them. There are many times when you need extra protection for your eyes. Safety glasses should be worn when carrying out some science experiments or when using power tools.

Light enters the eye through the *cornea* and passes through the 'hole' in the *iris*, called the *pupil*. The size of the pupil is controlled by the iris to vary the amount of light passing into the eye. Light then passes through the *lens*, which focuses the light rays to a sharp image on the retina, or back of the eye. The *retina* is packed with light-sensitive cells (*rods and cones*). These cells send electrical impulses to the brain through a bundle of nerve fibres called the *optic nerve*. There are no rods and cones present in the area where the optic nerve leaves the eye. This area is called the *blind spot*. Receiving a sharp image on the retina is not all there is to seeing objects; the brain has to interpret the messages it receives from the retina cells.

9 Study the diagram of the eye. Identify the parts of the eye numbered 1 to 10, from the list:



blind spot, cornea, iris, lens, muscles, optic nerve, pupil, retina, sclera, vitreous humour

Many devices have been invented to extend our senses, in other words, to allow us to observe things that we would otherwise not be able to observe. Lenses are used in microscopes, telescopes and binoculars to extend our sense of sight when objects are very small or far away. Lenses are also used to correct faulty vision, such as short-sightedness (*myopia*), long-sightedness (*hyperopia*) and some cases of cataracts. A cataract is when the lens of the eye or its capsule becomes opaque. Patients with cataracts have diminished vision, but can usually distinguish light from darkness. The most common form of cataracts (senile) are associated with aging, causing the whole

lens to become opaque. The most successful method of relieving senile cataracts is to remove the affected lens and replace it with an artificial lens.

10 What is meant by *opaque*?

Identify the following parts of the eye from their descriptions. (Questions 11 – 15)

- 11 Diaphragm that makes up the coloured part of the eye.
- 12 Sensory membrane that lines the inside of the eye.
- 13 Clear jelly that fills the inner part of the eyeball.
- 14 Nerve that carries electric pulses from retina to brain.
- 15 Dense white outer coat enclosing the eyeball excluding the cornea.
- 16 Name three cases when lenses can be used to aid the human eye.
- 17 Identify the glands that release tears.
- 18 Identify a use for tears in the eyes.
- 19 Identify two circumstances when eye protection should be worn.

Research

- 20 Find out what effect the 'blind spot' has on our vision.

Vision II

The *retina* is a thin sheet of cells lining the inside of the eyeball which contains photoreceptors. Photoreceptors are cells that contain light-sensitive pigments, such as rhodopsin. (A pigment is a substance that absorbs visible light of a specific frequency and reflects visible light at a different wavelength, ie has a definite colour.) These cells convert light images into electrochemical signals that are sent to the brain via the optic nerve for the brain to interpret. These photoreceptors are of two main types: rods and cones. In an average human eye there will be around 125 million rods and 6 million cones.

Rods are sensitive to blue-green light and are responsible for vision in dim light. They are more sensitive to light than cones, but cannot distinguish between colours. Rods are connected in groups and cannot distinguish fine detail.

Cones are responsible for colour vision. Colour blindness in humans is a result of certain colour-sensitive pigments being missing from cones. Colour blindness can be inherited and pedigrees can be used to describe how the condition is passed on. Cones can be individually connected to nerve fibres and give better visual acuity.

Nocturnal and diurnal animals have a different distribution of rods and cones, hence the difference in their night and day vision. The greatest concentration of cones (and no rods) is at the fovea (central area of retina). Concentration of rods decreases and concentration of cones increases towards the outer edges of the retina.

- 1 Give a definition of pigment.
- 2 Describe how cones and rods are located on the retina.
- 3 Which types of photoreceptors are responsible for colour vision?
- 4 How many types of cones are there in the human eye?
- 5 What is the role of rhodopsin in rods and cones?
- 6 Name the three colours to which the three types of rhodopsin in human cones are sensitive.
- 7 Describe the difference between *nocturnal* and *diurnal*.
- 8 Explain why cats have a large number of rods in their retinas.
- 9 Why might it be an advantage for animals that eat fruit to have colour vision?

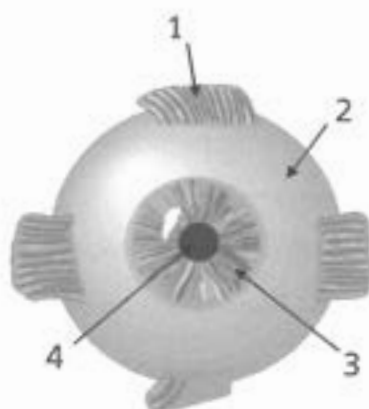
Genes are the units of inheritance and large numbers of genes make up chromosomes. Sex in humans is determined by X and Y chromosomes. Females are XX and males are XY. The X chromosome in humans carries the genes for colour vision. The commonest form of colour blindness is red-green, where either red or green sensitive cones are not present in the retina. This condition is many times more common in males than females. The gene responsible for red-green colour blindness is recessive. Females can be 'carriers' for the disease, ie they can have the gene for colour blindness on one chromosome but not the other, so the condition is not apparent but can be passed on. Pedigrees ('family trees') can be used to explain how colour blindness is inherited.

- 10 Give a definition of *chromosome*.
- 11 Why is red-green colour blindness described as a 'sex-linked' condition?

Using 'c' to denote colour blindness and 'C' to denote healthy genes, show genotypes for the following:

- 12 A male with normal colour vision.
- 13 A male with colour blindness.
- 14 A female with normal colour vision (2 possibilities).
- 15 A female with colour blindness.
- 16 Describe the conventions you used in representing genotypes.
- 17 Draw and explain a pedigree showing an inheritance pattern of colour blindness.

18 Study the illustration of the eye. Identify the parts: *iris, sclera/sclerotic coat, pupil, rectus muscle*.



- 1 _____
2 _____
3 _____
4 _____

Study the table and complete the missing information.

Eye part	Anatomy	Function
Conjunctiva	A mucous membrane lining the inner surface of the eyelids going over the forepart of the eyeball.	Lubricates eyeball.
Cornea	19	Allows light in and focuses it onto lens.
Sclera	Elastic coat enclosing eyeball, with exception of the cornea.	20
21	Layer of darkly pigmented tissue just inside the sclera.	Provides blood supply and absorbs light. Converts light to electrical signals that are sent to the brain.
Iris	Ring of pigmented tissue that can contract.	22
23	Transparent oval body suspended just behind the pupil.	Focuses light onto the retina.
Aqueous humour	24	Separates cornea and lens, gives shape to eye.
Vitreous humour	Transparent jelly behind lens filling the majority of the eyeball.	25
Ciliary body	Muscles embedded in the choroid attached to the lens by a ligament.	Contract to change shape of lens and focus light from different distances onto retina.
Optic nerve	26	Conduct electrical impulses from the retina to the brain.

Decide whether the following statements are true or false:

27 Colour blindness is a disease that is caused by an infection.

28 Stereoscopic vision is important in judging distances.

29 The shape of the lens can be changed by muscles inside the eye.

30 Myopia and hyperopia are forms of colour blindness.

31 There are three types of cones, each containing a separate pigment sensitive to either blue, red or green light.

The human ear

1 Use the following words to complete the passages that follow.

AUDITORY	BRAIN	CELLS	COCHLEA	EUSTACHIAN	HAIRS	HEAR
HERTZ	HIGH	MAGNIFY	MEMBRANE	NERVE	NEURAL	OSSICLES
OVAL	PINNA	SOUND	THREE	ULTRASONIC	BASILAR	

The human ear can be divided into _____ parts: the outer ear, middle ear and inner ear.

The _____ (auricle) is an outer flap of cartilage which collects sound waves and passes them into the _____ canal. At the end of this canal (approximately 3 cm in length) is the tympanic _____ or eardrum. When _____ waves strike the taut skin of the eardrum they cause it to vibrate. The middle ear has three tiny bones (_____) which vibrate when the eardrum does. The ossicles act as levers to _____ movement. The three tiny bones are the hammer (*malleus*), anvil (*incus*) and stirrup (*stapes*), named after the objects they resemble in shape. The amplified signal is conducted to the _____ window (*fenestra ovalis*) that connects the middle and inner ears. The _____ tube equals the pressure on both sides of the eardrum.

An important part of the inner ear is the _____, which resembles a snail shell and is filled with fluid. Alfonso Giacomo Gaspare Corti was an Italian anatomist who discovered (in 1850) the complex organ of the ear by which sound waves are converted to _____ impulses. The organ of Corti contains fine _____ that vibrate in response to sound waves. These hair _____ (located on the basilar membrane) vibrate according to their length. Short hairs vibrate in response to _____ frequencies and are located nearest the oval window. Those longer hairs at the end of the _____ membrane respond to lower frequency sounds. Each hair cell sends a signal to the cochlear (auditory) _____ which transmits the information to the _____.

Humans can only _____ a small range of sounds. The range of human hearing tends to be from 20 _____ to 20 kilohertz. Frequencies above 20 kHz are _____.

Complete the following table that shows the names, location and functions of parts of the human ear. The first row has already been done for you.

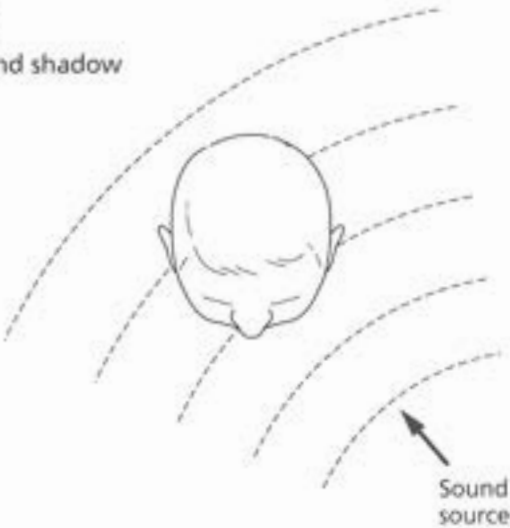
Part	Location	Function
External ear (pinna)	Outer ear	Directs sounds to auditory canal.
2	Outer ear	Takes sound to the eardrum.
Eardrum (tympanic membrane)	3	Vibrates when sound hits it.
Eustachian tube	Middle ear	4
5 (malleus)	Middle ear	Lever that magnifies movement due to sound.
Anvil (incus)	6	Lever that magnifies movement due to sound.
Stirrup (stapes)	Middle ear	7
8 (fenestra ovalis)	Inner ear	Connects to the throat using the Eustachian tube.
Cochlea	9	Turns vibrations to electrical signals.
Semicircular canals	Inner ear	10
11	Inner ear	Carries nervous signal from cochlea to the brain.

Decide whether the following statements are true or false:

- 12 Sound waves have kinetic energy.
- 13 The cochlea is filled with air.
- 14 The ossicles are made of cartilage.
- 15 The term *cochlea* comes from the Greek word meaning snail.
- 16 The organ of Corti is located in the semicircular canals.

The diagram shows a phenomenon called 'sound shadow'.

- 17 Referring to the diagram explain what is meant by sound shadow and the role it plays in the hearing of mammals.



Sound and communication

Sound is a form of energy that requires a medium for propagation, ie it can travel through a gas, liquid or solid, but not a vacuum. Vibrating objects produce sound. How quickly an object vibrates is called its frequency and is measured in cycles per second or hertz (Hz).

High notes have high frequencies and are produced by rapidly vibrating objects. Low notes have low frequencies and are produced by slowly vibrating objects. Pitch is the property of sound that we recognise as lower and higher notes.

The wavelength of a sound wave is related to frequency (and pitch). The higher the frequency, the shorter the wavelength. Wavelength is measured in metres.

- 1 Write these frequencies in words and numbers: 20 Hz 10 KHz
- 2 Give your own definition of *pitch*.
- 3 Using diagrams, show the relationship between wavelength and frequency for a low frequency sound and a high frequency sound.

The larynx, lungs, tongue and lips all play a part in how humans produce intelligible (understandable) sounds. The larynx ('voice box') is a hollow chamber near the top of the trachea ('windpipe') and contains the vocal cords. The vocal cords are two pairs of folded mucous membranes. The larynx is supported by cartilage (gristle) in places. Air passing from the lungs can cause the vocal cords to vibrate. The frequency of the sound is controlled by muscles that change the length and tension of the vocal cords. An analogy (comparison) can be made with the strings of a musical instrument that are plucked.

- 4 Give another name for the *voice box*.
- 5 Give another name for the *windpipe*.
- 6 Give another name for *gristle*.
- 7 Identify the following organ: *'The modified upper part of the trachea of air-breathing vertebrates that contains the vocal cords.'*

Animals

Apart from humans, many animals use sound to communicate. Male frogs can be very vocal. All frogs have a larynx that vibrates when air is forced through from the lungs. Male frogs of certain species have the lower part of the mouth as a large vocal sac (elastic pouch) used as a resonator to amplify their call. The sac inflates and vibrates like a drum. In some areas where frog population is high during the mating season, the noise produced by frogs can be in the region of 80 dB – about the same as a lawn mower.

- 8 Give three reasons why animals might communicate.
- 9 Give the full name for the unit *dB*.
- 10 Very high frequencies are used in 'echolocation' by some predators to find prey. Give an example of an animal that does this.
- 11 Very low sounds travel great distances under water and can be used by marine mammals to communicate. Give an example of an animal that does this.

The syrinx is the vocal organ (song box) of birds and is a special modification of the lower part of the trachea and/or bronchi.

Decide whether the following statements are true or false:

- 12 Humans are the only animals that use sound to communicate. _____
- 13 Vocal cords in humans are inflexible structures. _____
- 14 All sounds are produced by vibrations. _____
- 15 Birds generally use a vocal sac to make sounds. _____



Fish

Fish generate sounds with their swim bladders. Many fish have muscles that insert into the swim bladder. When the muscles are contracted, the swim bladder vibrates, producing a hollow sound. Some species, like the bull Atlantic Croaker (*Micropogonias undulatus*) make sounds at feeding times, while others make noise during reproductive periods.

Fish have good hearing but do not have external ears; sound waves are conducted by bones in the skull to the sensory chambers of the inner ear. Each chamber is lined with sensory hair cells. In some fishes, the inner ear is connected to the swim bladder that can resonate and 'amplify' sound. An external ear structure isn't needed because the density of fish bodies is similar to water, which means sound waves can pass through their bodies to the inner ear. Fish also have lateral lines. These sensory organs can detect low frequency vibrations and are used by the animal to swim in schools. Sound in water travels 4.5 times faster than in air and is felt as a pressure sensation in the sensitive lateral organs on fishes' sides as pressure waves.

Insects

Insects do not use voices to make sounds; often they use special structures on their bodies, wings or legs. In crickets, only the adult males chirp by scraping front wings together around 4000 times a second, to produce their musical noise. They do this to stake out territory or to attract mates.

Adult male cicadas produce a loud song using specialised sound-producing organs called *tymbals* (shaped like a kettle drum). Each species has a specific mate-attracting song. Female cicadas do not have tymbals, but in some species the females produce clicking or snapping sounds with their wings.

Insects don't have ears in the normal sense of the word. Insects can have 'ears' on their legs (frequently just below the knee), antennae or on the body. Many insects rely heavily on their hearing. Bees and caterpillars have sensitive hairs that detect vibrations, rather than sounds travelling through the air. Some theories suggest that ears of modern insects have evolved from vibration detectors.

Grasshoppers have six pairs of chordotonal organs (primitive stretch receptor-like hearing organs), along the sides of the body. Five sets of receptors lie along the side of the abdomen, and another more complex pair lies just above the hind legs. Some grasshoppers can hear calls (under certain conditions) from up to 2 kilometres away. A male mosquito can detect the 'buzz' of a female in flight by using antennae to pick up vibrations.

Ants don't seem to be sensitive to sounds travelling through the air, but can 'hear' with their knees, picking up vibrations. Many insects detect sound waves in air by using hairlike sense organs called trichoid sensilla, rather like the sound sensors in our ears. Some researchers claim that it is likely that ants could do the same, as their antennae are covered with such hairs.

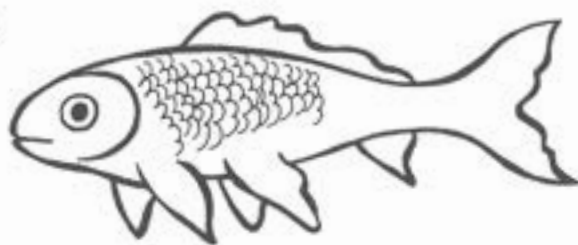
The sensitivity of those insects that can hear (butterflies, cicadas, crickets, flies and moths) is thought to extend into ultrasonic range, ie above the hearing range of humans, 20 kHz and above.

16 Draw up a table that shows all of the organisms mentioned in the information above, how they make sound and how they receive sounds.

17 What is meant by *ultrasonic*?

18 Why do fish not need external ears?

19 Does sound travel faster or slower in water compared to air?



The senses – definitions

Choose the correct words from the list to fit the definitions/descriptions.

ADRENALINE	AUTONOMIC	BINOCULAR	BRAIN	COCHLEA	CORNEA	CORTEX
EAR	EFFECTOR	ELECTRICAL	ELECTROMAGNETIC SPECTRUM	ENDOCRINE SYSTEM		
FOCUS	HORMONE	INVOLUNTARY	LENS	NEURONE	OPTIC	ORGANISM
PITUITARY	PUPIL	RECEPTOR	RETINA	SKIN	STIMULUS	TASTE BUDS

- 1 A living thing. _____
- 2 A change in outside conditions that makes an organism react. _____
- 3 Outer part of the brain that makes decisions and thinks. _____
- 4 The system of glands in the body that releases hormones. _____
- 5 Chemical produced by glands and carried through the blood. _____
- 6 A reflex or automatic action where choice is not involved. _____
- 7 Another name for a nerve cell. _____
- 8 Small bones in the middle ear. _____
- 9 A part of a living thing that receives signals. _____
- 10 The organ in the body that is concerned with sound. _____
- 11 Receptors associated with the tongue. _____
- 12 This has receptors that can detect heat, cold, pressure and touch. _____
- 13 Part of the eye that contains rods and cones. _____
- 14 The process by which the lens in an eye sees something clearly. _____
- 15 Along with the spinal cord, this makes up the central nervous system. _____
- 16 A fluid-filled tube in the inner ear. _____
- 17 The hormone that produces fright, flight and fight responses. _____
- 18 The gland that releases a growth hormone. _____
- 19 The type of impulses sent by nerve cells. _____
- 20 Part of the nervous system controlling heart beat, breathing and body temperature. _____
- 21 Hole in the iris that controls amount of light entering the eye. _____
- 22 To do with sight. _____
- 23 Vision with two eyes that gives an appreciation of distance. _____
- 24 An organ (eg muscle or gland) that physically responds to a signal. _____
- 25 The transparent front of the eyeball that covers the iris and pupil. _____
- 26 Part of the eye that brings light rays to a focus on the retina. _____
- 27 Visible light is part of this. _____

The senses review quiz

Name: _____

Part A: Multiple choice

Choose the best answer for each question, and write the corresponding letter in the boxes provided.

1 The correct sequence describing receptors responding to stimuli is:

- | | | | | | | |
|---|----------|-----------|-----------|----------|-----------|--------------------------|
| A | receptor | effector | messenger | stimulus | response | |
| B | receptor | response | effector | stimulus | messenger | |
| C | stimulus | messenger | effector | receptor | response | ☐ |
| D | stimulus | receptor | messenger | effector | response | |

2 Parts of the body that detect a stimulus are called:

- A receptors
B stimuli
C messengers
D effectors

☐

3 In a vacuum, electromagnetic waves carry energy in straight lines at a rate of:

- A 200 000 km/sec
B 300 000 km/sec
C 3 000 000 km/sec
D 200 000 km/sec

☐

4 Visible light is referred to as:

- A white light
B blue light
C red light
D yellow light

☐

5 An example of a stimulus is:

- A ear
B light
C nose
D blinking

☐

6 An example of a receptor is:

- A eye
B touch
C light
D cold

☐

7 The type of vision resulting when both eyes are positioned at the front of the head is:

- A monocular
B double
C biocular
D binocular

☐

8 Visible light has wavelengths between:

- A 200-400 nm
- B 400-700 nm
- C 700-1000 nm
- D 100-200 nm

☐

9 Wavelength describes:

- A length from one wave peak to a trough
- B the measure at the highest peak of a wave
- C the measure at the lowest trough of a wave
- D the distance from peak to peak, or trough to trough, of a wave

☐

10 Visual acuity describes:

- A the shape of the eye
- B the ability for special glands to moisten the eyes
- C how sharply the eye perceives objects
- D the position of the eyes on the head

☐

11 Thermoreceptors respond to which of the following stimuli?

- A light
- B chemical
- C heat
- D mechanical

☐

12 Which does not refer to a part of the ear's structure?

- A ossicle
- B cochlea
- C neural tissue
- D pinna

☐

13 Which part of the human body contains the organ of Corti?

- A eye
- B brain
- C ear
- D tongue

☐

14 The 'window' that connects the middle ear and inner ear is:

- A round
- B oval
- C square
- D triangular

☐



Part B: True or false

Indicate whether the following statements are true or false.

- | | |
|--|-------|
| 15 Vibrating objects produce sound. | _____ |
| 16 Fish use swim bladders to make sound. | _____ |
| 17 The higher the sound frequency, the shorter the wavelength. | _____ |
| 18 The property of sound we recognise is called 'pitch'. | _____ |
| 19 Butterflies cannot hear. | _____ |
| 20 Ants can hear with their knees. | _____ |
| 21 Sound is a form of energy. | _____ |
| 22 The trachea is sometimes called the 'windpipe'. | _____ |
| 23 Sound can travel through gas. | _____ |
| 24 Frogs lack a larynx. | _____ |
| 25 Low sounds travel short distances in water. | _____ |
| 26 A <i>syrix</i> is a type of voice box. | _____ |
| 27 The larynx is also called a 'voice box'. | _____ |
| 28 Insects that can hear, do so at a lower ultrasonic range than humans. | _____ |

Part C: Definitions

Identify the terms described in the following definitions.

- | | |
|---|-------|
| 29 The part of the human eye responsible for colour vision. | _____ |
| 30 Thin sheet of cells containing photoreceptors lining the inside of the eyeball. | _____ |
| 31 Specialised nerve cells: transmit signals by electro-chemical changes in their membranes. | _____ |
| 32 The brain and the spinal cord. | _____ |
| 33 A hollow chamber near the top of the trachea that contains the vocal cords. | _____ |
| 34 Billions of neurones, interconnecting with each other that control our actions and feelings. | _____ |
| 35 Eye condition where image is focused in front of the retina. | _____ |
| 36 Bending of light as it passes between different media. | _____ |
| 37 Communication involving the eye. | _____ |
| 38 The ability of the eye to focus on objects at different distances. | _____ |
| 39 Caused by the lack of one or more colour-sensitive pigments in the cones of the eye. | _____ |
| 40 Tube in ear that equalises pressure. | _____ |

Historic figures in genetics

Mendel

You will no doubt have heard of the botanist-monk Gregor Mendel (1822 – 1884) and his famous pea experiments. Mendel's observations of the different traits expressed by pea plants led to recognising that each parent contributed heritable 'units' or 'particles' to their offspring.

Mendel studied seven characteristics of garden peas. Two of the most obvious contrasting characteristics he used were the shape of seeds and the length of stems. He hand-pollinated the pea plants.

Mendel used statistics to help him realise that inherited characteristics are paired units. When he crossed pure breeding tall plants with pure breeding short plants, all of the first generation (F_1) offspring were tall. When the F_1 generation were crossed with each other, the resulting second generation (F_2) had a ratio of three tall plants to every short plant.

- 1 Define *trait*.
- 2 Why would Mendel go to the bother of hand-pollinating pea plants when nature would have done it for him?
- 3 Define *statistics*.
- 4 Give the genetic definition for *pure breeding*.

Mendel was unable to identify exactly what, or where, those units of inherited characteristics were found but the realisation that the 'units' had to exist was a huge step forward in science. Prior to Mendel introducing this concept, people had to rely on *morphology* and *taxonomy* when trying to work out species relationships. We now recognise Mendel's 'units' as genes, the basic units of inheritance, but it took time to narrow down exactly what genes were and where genes were located.

- 5 Which branch of science does this define: '*The study of the form and structure of organisms*'?
- 6 Which branch of science does this define: '*The classification of plants and animals*'?
- 7 Suggest possible problems faced by using only physical characteristics to predict parent/offspring relationships.

Mendel came up with two main principles. His first principle, the 'law of segregation', stated that somatic (body) cells have pairs of genes (although he didn't use the word 'genes'), but segregate, or separate, when sex cells (eggs or sperm) form. In the process, each member of the pair becomes part of the gamete (sex cell). Mendel's contemporaries believed that physical characteristics 'blended' when they were passed from one generation to the next. Mendel's research showed him that this was not generally true, but that hereditary 'units' were passed on relatively unchanged. This led to his second principle, the 'law of independent assortment', which states that the phenotype (physical characteristic caused by a gene) for any single characteristic is usually not influenced by the phenotype for another characteristic. Since his laws became widely known (many years after his death) they have formed the foundation for modern genetics and heredity.

- 8 Define *gamete*.
- 9 Define *phenotype*.
- 10 In your own words explain what is meant by Mendel's first principle.
- 11 In your own words explain what is meant by Mendel's second principle.

Research

- 12 Write a report on the major achievements of one of the following pioneers in genetics research:

a Watson & Crick	e Nirenberg, Ochoa & Khorana
b T.H. Morgan	f Boyer, Cohen & Berg
c Avery, MacLeod & McCarty	g Reginald Crundall Punnett
d Hershey & Chase	h Barbara McClintock

Sutton and Boveri

History has many instances of simultaneous discoveries made by scientists working independently. An example occurred in 1902 when American geneticist Walter Sutton (1877–1916) and German scientist Theodor Boveri (1862–1915) both concluded that genes had to be located in the cell's nucleus. Sutton and Boveri reached this conclusion using the logic that to produce a zygote, parents make equal contributions in the form of egg and sperm. Since both contribute equally, whatever the sperm and egg shared in common, in terms of content, had to be where the genes were found. Eggs contain many components absent in sperm. In fact sperm is little more than a nucleus full of genetic material and a tail with which to deliver it. Chromosomes are the main component of the nucleus so the scientists rightly concluded that the genetic material must be located on chromosomes.

Sutton put forward the idea that genes were found on the chromosomes partially as a result of his observations of meiosis.

Morgan

Geneticist Thomas Hunt Morgan (1866–1945) originally doubted the theory that genes were found on chromosomes. He was one of many scientists who believed that genes were separate units and not part of a larger structure. Interestingly, his 1910–1911 research ended up confirming that they were.

13 Shown here is a simplified diagram of a pair of chromosomes.

On the diagram label the following:

- a alleles of a gene
- b centromere
- c chromatid

14 What does this describe: 'The part of a chromosome to which the spindle attaches during mitosis and meiosis'?

Decide whether the following statements are true or false:

15 Mendel invented the term 'gene'.

16 A zygote consists of a sperm fused with an egg.

17 Genes are found on chromosomes.

18 Sperm have the same components as ova.

19 Chromosomes are usually found in the cytoplasm of a cell.

20 TH Morgan used a variety of fruit fly in his genetics experiments because fruit flies breed very rapidly.

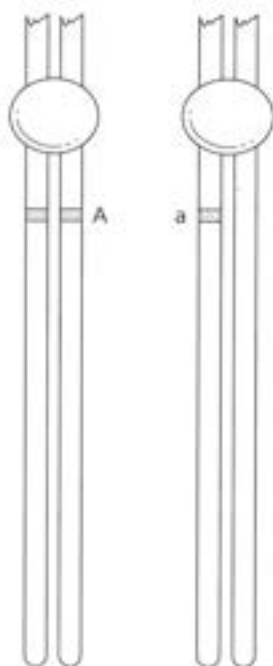
21 Sex cells are also called somatic cells.

22 Meiosis is the process that doubles the number of chromosomes in sex cells.

23 Chromosomes are made up of many genes.

24 Inherited traits are controlled by factors that occur in pairs.

Only part of the chromosomes shown



Review of genetics terms

A thorough knowledge of the specialist terms used in Genetics is essential for an understanding of the subject. Read this page carefully. Using the information given and a separate text book (or other source) write your own definitions for the terms shown in *italics*.

The *gene* is the basic unit of inheritance for a given trait. *Alleles* are alternative forms of a gene. For example genes can code for eye colour and the alleles include varying shades of brown, blue, green and even violet.

A *dominant* allele produces a trait that overrides the presence of another trait (*dominate* to control or rule). A capital (upper case) letter indicates dominant traits. For example, brown eyes are a dominant trait over blue eyes. If an individual receives a brown (B) allele from its mother and a brown from its father, the genotype is BB and brown eyes would be present. If the same individual received a brown allele from its mother and a blue from its father, the genotype would be Bb and it would still have brown eyes. The only way for the blue trait to appear is if it is the only one present (bb). This is because the blue allele is *recessive* (meaning to recede or go back). A small (lower case) letter indicates recessive traits.

Homo means 'the same' and *homozygous* allele pairs are the same, such as BB (meaning two dominant, brown alleles) or bb (two recessive blues). *Hetero* means 'different' and *heterozygous* allele pairs contain different alleles, such as Bb.

The *genotype* refers to the complete genetic make up of an individual whether or not it is expressed. The *phenotype* describes what we see (from a Greek version of *pheno*, meaning 'what is seen'). An individual may carry a Bb eye colour genotype. In other words one brown and one blue eye colour allele. However, the phenotype will be brown. The brown eyes do not tell us if the individual is homozygous dominant (BB), meaning that both dominant alleles are present, or heterozygous dominant (Bb), meaning two alleles are present but only the dominant one is being expressed.

As the name indicates, *lethal* alleles can kill. Some lethal alleles need only be present in one dose to kill. In this case they are called *dominant lethal*, while in others a double dose is required (*recessive lethal*). *Conditional lethals* are alleles that will only be lethal under certain conditions. For example, some will only be activated in higher than normal temperatures. If the organism carrying the conditional lethal is not exposed to the higher temperatures, it remains unharmed. Lethal alleles are not usually found in high frequencies in populations though, as always, exceptions occur.

There are relatively few *chromosomes* in the nucleus of a cell of an organism. Because there are so many inheritable characteristics, each chromosome must consist of many genes. A chromosome consists of a *centromere*, a circular structure, and paired strands called *chromatids* attached to the centromere.

The term *autosome* refers to all chromosomes that are not sex chromosomes. *Sex chromosomes* are those that determine the sex an organism will become. Lethal genes can occur on autosomes or sex chromosomes. The term *autosomal lethal* tells us that the lethal gene is located on a non-sex chromosome. The frequency of *sex-linked lethals* can vary. If the lethal is dominant, only one dose is required to kill so either sex can expect fairly equal ratios.

If the lethal is *X-linked recessive* the distribution will differ between the sexes. This is because females receive two X chromosomes and would therefore require a double dose of the recessive for it to be expressed (therefore lethal) whereas males, who receive one X and one Y, would be affected by a single dose. As the Y chromosome carries very few genes, sex-linked lethals are also called *X-linked* because they are found on the X chromosome.

Punnett squares, peas and crossing

Punnett

Reginald Crundall Punnett (1875–1967), together with William Bateson, discovered genetic linkage, sex determination, sex linkage and autosomal linkage, using poultry and sweet peas.

The prefix *mono* means 'one' and a *monohybrid cross* is a cross that occurs for a single trait, such as the texture of a pea's outer coat. Mendel observed that the coat can be wrinkled, a recessive trait, or round, a dominant trait. If we consider the wrinkled/round trait, the alleles could be shown as (r) for wrinkled and (R) for round. (The convention is to represent the recessive allele using lower case initial and the dominant allele as an upper case initial). A Punnett square (figure 1) can be drawn to represent a cross between a homozygous round pea (RR) and a homozygous wrinkled pea (rr). The offspring (F_1 generation) are all heterozygous (Rr) peas.

	R	R
r	Rr	Rr
r	Rr	Rr

Figure 1

- 1 Define *homozygous*.
- 2 Define *heterozygous*.
- 3 Use the blank Punnett square (figure 2) to show the genotypes resulting from a cross between (Rr) F_1 offspring.
- 4 Describe the phenotypes (physical characteristics) of the resulting offspring.
- 5 What is the ratio of the phenotypes described in Question 4?
- 6 What is the general name for the offspring of the F_1 generation?

Figure 2

Dihybrid crosses (*di* two) are crosses for two traits, such as the texture and colour of the pea's outer coat: wrinkled (r) or round (R) and yellow (g) or green (G). Dihybrid Punnett squares are done in much the same way as monohybrid crosses, except that you are working with two traits. Imagine a cross between seeds with a genotype of RrGg. The possible gametes are RG, Rg, rG and rg. Figure 3 shows all the genotypes for the RrGg x RrGg cross.

- 7 In the space below each cross in figure 3, write the resulting phenotype for each individual genotype.

	RG	Rg	rG	rg
RG	RRGG	RRGg	RrGG	RrGg
Rg	RRGg	Rrgg	RrGg	Rrgg
rG	RRGG	RrGg	rrGG	rrGg
rg	RrGg	Rrgg	rrGg	rrgg

Figure 3

Role and structure of chromosomes

Chromosomes are housed in the cell's nucleus. Chromosomes contain a nucleic acid called DNA, or deoxyribonucleic acid (deoxy means 'less oxygen'). DNA carries an organism's genetic information*. The many tiny units that make up the DNA molecule are genes. A gene is a series of nucleotides and contains specific information about which characteristics the organism will inherit.

* DNA also has assorted proteins attached to it.

The gene is the smallest unit of heredity and genes are linked by a sugar-phosphate 'backbone'. The name given to the unit that is formed by the deoxyribose, the phosphate and a nitrogen base, is *nucleotide*. Weak attractions ('hydrogen bonding') between strands cause them to twist into a double helix shape (think of a spiral staircase). The whole unit is called a chromosome.

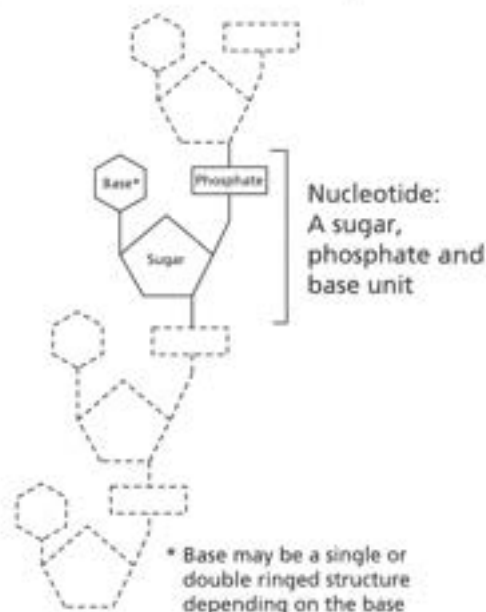
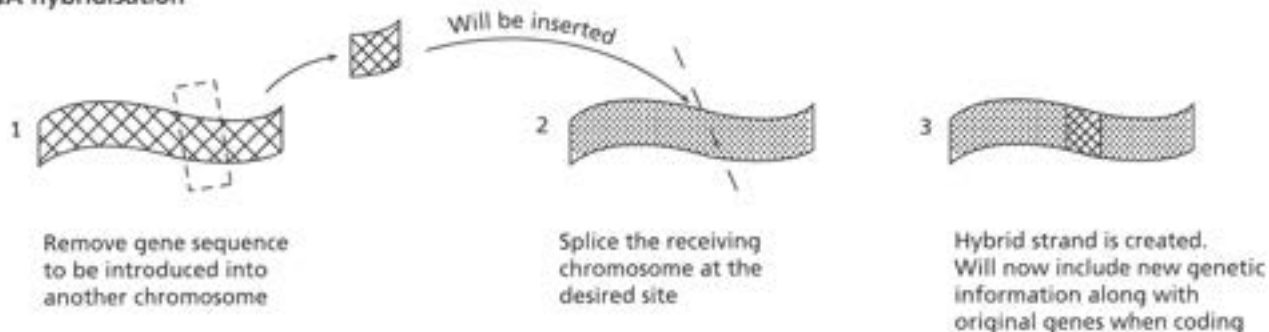
- 1 *Drosophilla melanogaster*, the common fruit fly, has four chromosomes. How many did it receive from its mother?

Genetic manipulation:

How humans can interfere with genetic material

Genetic manipulation has been occurring for some time now and, despite protests, it will continue. Many techniques used for genetic manipulation depend on the DNA's ability to repair itself (to a degree) should it become slightly damaged in the process.

DNA hybridisation



Technically, a hybrid is a cross between two genetically different organisms. Hybrid DNA is created by breaking (denaturing) the protein bonds on the strand without completely destroying them. (Consider them loosened.) Heat is used to denature proteins and a slight elevation in temperature can cause bonds to loosen and break but, if handled carefully, they can reform. In this way unrelated pieces of genetic material can be joined to a strand of DNA. This technique can be used to create some forms of genetically modified organisms (GMOs).

- 2 In the USA, about 39% of crops contain some genetically modified plants, whereas in China it is 2% and India it is 0.1%. Discuss why this might be the case.
- 3 Define *nucleotide*.
- 4 What causes the two strands in DNA to form the double helix shape?
- 5 Define *denatured*, with respect to DNA.

What do the following terms define?

- 6 *Nucleic acid located in nucleus of cell, which is the basis of heredity.*
- 7 *The deliberate alteration of the DNA inside a cell's nucleus with the aim of modifying that organism.*
- 8 *Producing a genetically identical duplicate of an organism.*
- 9 *Cell body (made up of genes) that plays a large role in inheritance.*

Genetic issues in the 21st century

Many of you will be familiar with the term *genethics*, or the ethics (morals) behind genetic research. Since 2001, a few new terms have arisen, two of which you will probably hear at some time. These are *genomics*, a word that refers to an organism's genome (complete genetic make-up) and *proteomics*, which refers to the complete set of proteins found within an organism – and there are many!

It is just as important to understand and map the human proteome as it was to map the human genome – some researchers might argue it is more important. At one time, people believed that each protein was responsible for one, possibly two, tasks within the body. We now know that a single protein can be involved in over 1000 tasks. This is of particular concern to pharmaceutical researchers. Many current pharmaceuticals are designed to target and affect one or two proteins at most. Knowing that proteins work collectively, and in complex ways, revealed the importance of re-thinking current methods of pharmaceutical development.

1 Define *pharmaceuticals*.

Barbara McClintock: Founder of cytogenetics

American scientist Barbara McClintock (1902–1992) provided indisputable, physical evidence that genes are transposable. Her work proved that genetic material was exchanged between matching chromosomes and the shuffling of genes produced the variation we recognise between related organisms. Barbara McClintock, a student of Thomas Hunt Morgan, is reported to have endured considerable ridicule from people who thought she was wrong. In the 1930s, with the assistance of Harriet Creighton, she set about providing evidence for her theory. They used maize to show how genetic recombination occurs when chromosomes cross over each other, break and recombine their genetic material to produce new combinations. For her research, Barbara McClintock was awarded the 1983 Nobel Prize for Medicine and Physiology. She was the first woman in America to win an unshared Nobel Prize.

2 Define *transposable*.

Implications of gene technology

Gene technology is not new but its implications continue to generate debate. Genetically modified (GM) foods worry many people because there has not been time to assess the long-term effects of ingesting them. Some of the main aims of using gene technology include:

- *improving foods (plant or animal)*: pest and disease resistance, addition of vitamins and other substances currently deficient in millions of diets, increasing growth time (and in some cases size).
- *producing medicines in new ways*: for example, modifying certain crops to produce substances they did not produce before, such as developing potato crops to produce a protein vital to create an oral hepatitis B vaccine.
- *treating diseases resulting from defective genes*: for example, replacing defective genes with functional ones (a hybridisation technique).

3 What is meant by an *oral vaccine*?

4 Give one reason why some people are against GM foods.

5 Give one reason why some people are for GM foods.

Give definitions for the following terms:

6 *genethics*

7 *genomics*

8 *proteomics*

9 *hybridisation*



At present *selective breeding* is used to produce plant and animal crops with desired features. Selective breeding directs the 'evolution' of a species by using only organisms that possess the desired trait(s) as breeding stock. In some instances, crossbreeding is used to do this. *Crossbreeding* uses closely related organisms that may not breed under natural circumstances but are still considered 'acceptably related' for the purpose.

Where gene technology differs drastically from this is that it can introduce genetic material from completely unrelated species. The 'mix and match' is made when researchers believe that an organism, if it possessed the ability to produce a given substance, could use this modification in their own genetic make-up to produce a new result.

How is the DNA modified?

Scientists identify and isolate the strand of DNA on which a required trait is found. The DNA is then replicated many thousands of times producing clones of the original strand. The area in which the required genes are located is identified and the required piece is spliced from the cloned strand. The spliced piece of DNA is inserted into the DNA of another organism. DNA is responsible for protein development and proteins instruct certain reactions to occur. Thus the introduced DNA will produce changes in the reactions that occur in the future.

This technology has been used to instruct the genes of ewes to produce substances in their milk, which will be extracted for the treatment of diseases such as cystic fibrosis. In some instances, gene therapy will be used to splice defective genes from an individual where it will be replaced by properly functioning genes (eg muscular dystrophy). Since it is impossible to replace every defective gene in a body, particularly in adults, some therapies will be used to clone many thousands of 'repaired' DNA, which will then be reintroduced into the individual, providing enough effective coding to correct (at least partially) the problem. Hopefully, in the case of developing embryos, for some conditions, defective DNA can be repaired or replaced before the child is born.

Give definitions for the following terms from the information above:

- | | |
|-----------------------|--|
| 10 selective breeding | 15 ewes |
| 11 crossbreeding | 16 therapies |
| 12 isolate | 17 Give one example of <i>selective breeding</i> . |
| 13 replicated | 18 Give one example of <i>crossbreeding</i> . |
| 14 spliced | |

Research

- 19 There will always be a place for *traditional* and *organic* farming methods.
Write brief definitions for each method. Include an explanation of their similarities and differences.
- 20 In the box provided below, sketch the basic steps involved in DNA modification.



A pedigree table (family tree) is often used in genetics to show how characteristics are passed from one generation to the next. A pedigree is usually set out with the earliest known ancestors at the top of the table and succeeding generations in line of descent including direct and collateral descendents. A pedigree is especially useful in tracing the transfer of sex-linked characteristics. There are many conventions used in drawing up these tables.

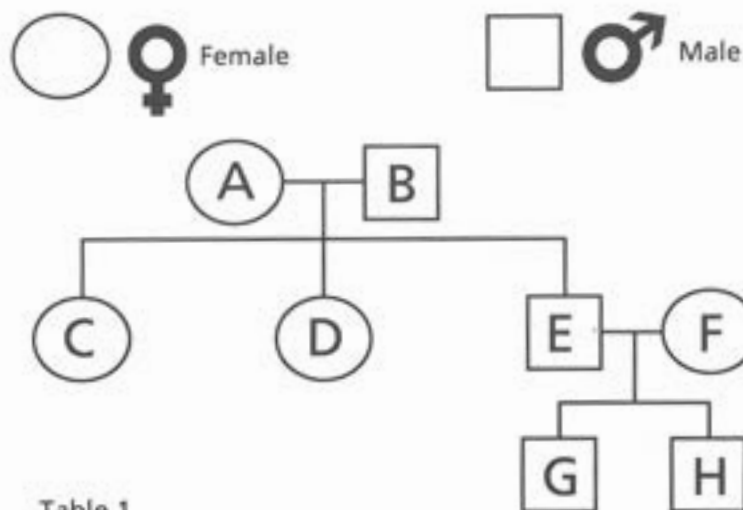


Table 1

Table 1 shows a particular human family. Female A marries male B and has three offspring, C (female), D (female) and E (male).

E marries female F and this couple have two male offspring, G and H.

1 Use a similar system to draw up a pedigree for the following family:

A male and female married and had two offspring, one boy and one girl. The male never married and had no offspring. The female married and had three children, two boys and one girl. All of these children subsequently married. The first male of this generation had two female offspring, the second male had no offspring and the female had one boy and one girl.

2 Explain the difference between *direct* and *collateral* descendents.

Phenotypes are the physical characteristics resulting from inherited genotypes. Pedigrees can be used to illustrate how physical characteristics occur in generations of a family.

3 What is the name of the technique used to predict possible genotypes and phenotypes of offspring?

4 Explain what is meant by F_1 , F_2 and F_3 when referring to generations.

Table 2 shows the occurrence of lobed ears in a family.

5 Describe what is shown by this table.

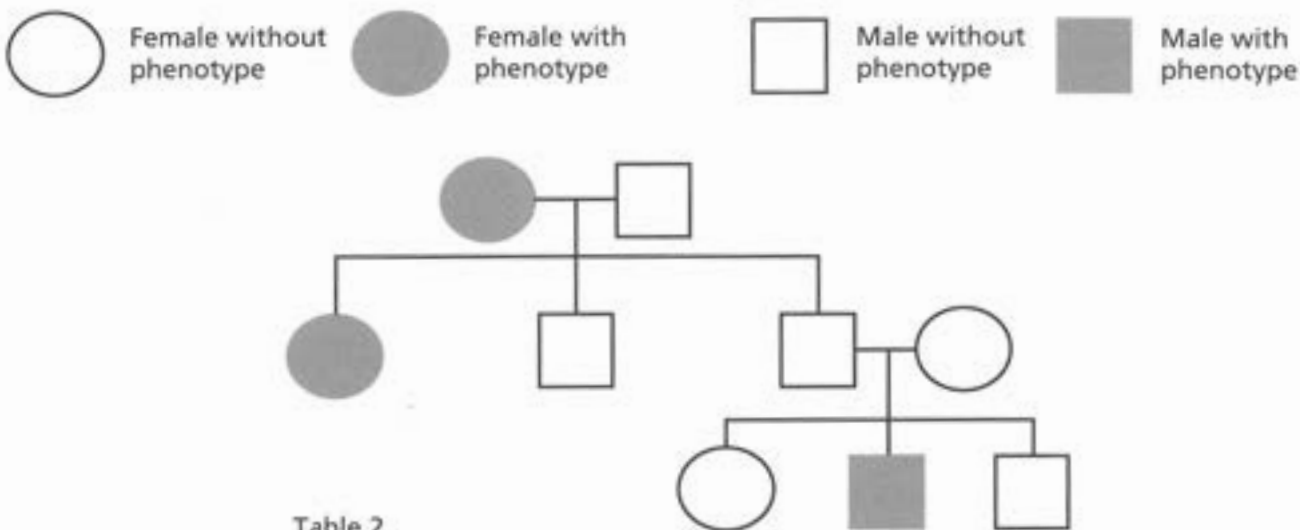


Table 2

Sex-linkage

Humans characteristically have two sex chromosomes. Males are XY and females are XX. The Y chromosome is smaller, has a different shape to the X chromosome and carries many fewer genes. The Y gene lacks partners for many of its genes. Genes that are unpaired (ie exist only on the X chromosome) are sometimes referred to as *sex-linked genes*. Many inherited defects are due to recessive genes.

A 'carrier' is a person who is carrying a recessive gene for a disease but is not affected by that condition. If the dominant gene is 'normal', it overcomes the recessive gene. Because the Y chromosome has 'missing' genes, sometimes there may not be a dominant gene present and the characteristic carried by the recessive gene may show up. It is possible for a male to suffer from a sex-linked disease by inheriting a single recessive gene (on the X chromosome) for that condition from his mother.

- 6 Explain the difference between *dominant* and *recessive* genes.
- 7 Why is it less likely for a female to suffer from a disease due to faults in the X chromosome?
- 8 A carrier for a disease is likely to be heterozygous for that disease. Define *heterozygous*.

Sex and gender

The terms *sex* and *gender* are often confused. Sex refers to how we differentiate organisms by the part they play in the reproductive process. We usually refer to the organism that donates genetic information as male and the organism that receives genetic information as female. Even bacteria can be divided into male and female on the basis of how they swap chromosomal material. More complex organisms have *gonads* – reproductive organs that differ between males and females. *Hermaphrodites* are organisms that possess both male and female reproductive systems. In mammals, true hermaphrodites are rare. Generally, one of the reproductive systems is vestigial. Most hermaphrodites are sterile, however, some sperm and eggs can be produced (far less sperm). Many of the gametes produced are unviable though births have occurred. More than half of human hermaphrodites are raised as males. Some organisms are also capable of *parthenogenesis* (Gk virgin birth) where unfertilised ova can grow into adults genetically identical to the sole parent.

Gender is a society-based classification of males and females. This could be the expectation that (in some cultures) women wear skirts and have long hair, raise children etc and men wear trousers, have short hair and go out to work. Different societies will have different ideas (often stereotypes) as to what are 'male' and 'female' roles.

What do the following terms define?

- 9 An organ that makes gametes (sex cells).
- 10 An organism that has both ovaries and testes.
- 11 The development of an organism from an unfertilised gamete.
- 12 Give two characteristics that differentiate males and females by sex.
- 13 Give two characteristics that differentiate males and females by gender.
- 14 Define *vestigial* and *unviable*.
- 15 Suggest why many hermaphrodites are raised as males.

Some physically healthy males and females experience other gender identity issues. These can arise when an individual feels mentally like the opposite sex, or are confused as to which sex they were meant to be or which gender they feel themselves sexually attracted to.

Give definitions for the following gender recognition issues:

- 16 Homosexual
- 17 Transvestite
- 18 Transsexual



Use words from this list to identify the terms.

ALLELES	CHARACTERISTIC	CHROMOSOMES	CLONING	CO-DOMINANCE	
DIPLOID	DOMINANT	DOUBLE HELIX	EMBRYOLOGY	FERTILISATION	GAMETE
GENETIC ENGINEERING		HETEROZYGOUS	INHERITANCE	MEIOSIS	MUTATION
NUCLEOTIDE	HYBRID	PEDIGREE	POLYPEPTIDE	PUNNETT	RESOURCES
	EVOLUTION	REPLICATION	SUTTON	TRANSGENIC	

- 1 Change of organisms over time.
- 2 Used by organisms to maintain life.
- 3 The study of developing embryos of different species.
- 4 Distinguishing feature of an organism.
- 5 Process by which we receive characteristics from an ancestor.
- 6 Having different alleles for a given trait.
- 7 A pair (or more) of genes that determine a specific characteristic.
- 8 Genes that will be expressed in a phenotype.
- 9 Table showing family tree.
- 10 Humans have 23 pairs of these – contain genetic information.
- 11 The symbol (2n) means two chromosomes (paired). The word for 2n is:
- 12 Cell division that produces sex cells.
- 13 Sex cell.
- 14 When male and female sex cells fuse.
- 15 Neither completely dominant nor completely recessive.
- 16 Building block that makes up DNA strand.
- 17 Contains two or more amino acids.
- 18 Sudden change in DNA.
- 19 Technique used to alter the genetic makeup of an organism.
- 20 Asexual reproduction resulting in genetically identical organisms.
- 21 Organism that has received genetic material from a different species.
- 22 Along with Boveri, identified the importance of chromosomes in inheritance.
- 23 Model of DNA proposed by Crick, Franklin and Watson.
- 24 Squares used to predict inheritance of phenotypes and genotypes.
- 25 Process of duplicating the genetic material within a nucleus.
- 26 Produced when the genetic material of two species is mixed.

Medical ethics

Ethics: The science of morals. A society's obligations and responsibilities.

Whether discussing child raising, politics, farming or rules and regulations, we all have opinions about what is 'right' and what is 'wrong', what is 'good' and what is 'bad'. The medical community is not excused from being expected to follow acceptable procedures with regard to their practices.

There is no single set of international rules because what is considered acceptable in one country can be considered unacceptable in another. For example, in Western countries, like Australia, an adult will discuss their ailment and possible treatment with their doctor. The doctor will only invite family members into the discussion with the patient's consent. In many Eastern countries, such as China, the patient's condition is discussed with the whole family – in particular the elder relatives – and decisions are made as a group. The patient is not necessarily given a choice about this.

- 1 State two advantages and two disadvantages for the 'Western' and 'Eastern' ethics when it comes to discussing a patient's condition.

Many people have argued that teaching ethics to members of the scientific community is a waste of time. They believe technology is moving at such an alarming rate that the ethical issues can't possibly be imagined and often, once they are realised, it is too late to act. However, most will agree that it is important to have a standard of practice that should always be considered and that using these guidelines, individual cases can be measured against them.

Doctors are now being placed in difficult positions. Many years ago euthanasia, genetic engineering, IVF and stem cell research were unknown. Today, these present complex ethical issues. For some people, any kind of modification to the human body is unacceptable – '*we shouldn't play God*'. Culture and religion play important roles in ethical considerations. For some religions, blood transfusions are never permitted (even if a life would be saved) whereas others believe that if humans were given the gift of learning how to do these things, they should be done provided they would improve the quality of life of the patient.

Examples of contemporary ethical issues include:

- '*designer babies*': selecting embryos with characteristics the parents want
- the rights of the unborn
- gene technology: genetic modification, genetic screening, cloning etc
- organ transplants
- the treatment of living subjects for experiment trials (humans and other animals)
- euthanasia: '*humanely*' killing a person '*before their time*' (*eu* well, *thanatos* death)
- involuntary sterilisation, eg to prevent a mentally impaired individual from becoming a parent
- surrogate mothers: '*substitutes*' who carry a developing baby
- sperm and egg donation and IVF techniques

Genetic screening

In Australia today, most pregnant women are given the opportunity to choose whether they want to screen for the presence of a genetic disorder in their developing child. If the genetic screening does identify a disorder, several things must be considered. These include: how badly affected will the child be? If it is mild disorder, will the child be able to live a normal or near-normal life? What are the costs involved? Will expensive treatments and/or equipment be required? How will the rest of the family be affected?

Once a disorder is identified, parents must consider its severity. A mild disorder will present a different set of circumstances than a severe one. The long-term and short-term considerations must also be noted. If a baby will be born with a severe condition it may die young. This means the parents will be faced with the additional trauma of the death of a child. Alternatively, if the child can live to a fairly old age but will be severely impaired, is it fair to let them (and the rest of the family) suffer?



2 Many people disagree with testing a developing embryo to determine if it is carrying a potentially disabling condition. Do you agree or disagree with this? Why?

3 When tests mentioned in question 2 show the presence of a genetic disorder, the parents could be faced with other ethical issues. State one other possible ethical issue that could arise.

A developing embryo tests positive for a genetic disorder such as Cystic Fibrosis. The father's side of the family says the child has the right to live. The mother's side disagrees saying it is unfair to put the family through the heartache of watching their child suffer. They also argue that the child doesn't deserve to 'exist' in such a condition, knowing that its siblings are all healthy and 'normal'.

4 What other issues must the family described above have to consider?

5 In the example above which side would you support, the mother's family or the father's? Why?

'The welfare of the patient is the doctor's first and only concern. In the case of a pregnant woman, the doctor cannot possibly be expected to put the welfare of the developing baby before his patient.'

6 Discuss this statement. Do you agree or disagree? Give your reasons.

7 With reference to question 6, when might the welfare of a developing baby become a priority?

Scientific research

Whenever new scientific research is undertaken many things must be considered before the project can proceed. Ethics committees, made up of a broad range of the community (scientists, religious etc) are involved in the decision making once the 'science' has been mapped out in theory. Considerations include:

- What is the purpose of the procedure?
- If live experimental subjects are required (of any species, including human) will any suffering be involved? If so, can the researchers prove that the experiments cannot possibly be done any other way and that they will use the minimum number of subjects to obtain valid results, eg will not cause suffering to more subjects than is necessary?
- Will the procedure/experiment work in practice as well as in theory?
- How will it be done? Full details to be given: costs, methods, length of time etc.
- It is ethically acceptable? (A particularly hot issue when suffering will be involved.)
- Who will pay for it? (taxpayer, business community, government etc)

8 Do you consider it ethical for the taxpayer to fully or partially fund expensive treatments and/or equipment (such as specialists, wheelchairs etc)? Explain your choice.

Jenner and vaccination

Edward Jenner (1749–1823) was a country doctor in the west of England. He is famous for his work on smallpox. This disease was a major killer of infants and young children in the eighteenth century. He noticed that people who had suffered the (relatively) mild disease cowpox did not catch smallpox. He experimented, putting some pus from cowpox blisters from a milkmaid into a young boy and later injecting the boy with smallpox. The boy was ill for a few days but fully recovered. This process of *vaccination* (Latin *vacca*, a cow) immunised against smallpox.

9 Discuss the 'rights' and 'wrongs' of Jenner experimenting with a live subject.

In vitro fertilisation

Not all couples that wish to have a baby are able to do so. About 15% of couples will experience difficulty in trying to achieve a pregnancy. There are many reasons why a couple can be considered 'infertile', and sometimes no cause for the infertility can be found.

- 1 Give two reasons why a male might be infertile.
- 2 Give two reasons why a female might be infertile.

The infertile couple has options. One of these is *in vitro fertilisation*, or *IVF*, a medical procedure to help fertilisation. *In vitro fertilisation* – literally, 'in-glass' fertilisation – involves fertilisation of ova outside of the body. Techniques vary, but the following shows a common sequence:

- Fertility drugs are given to the female to stimulate ova production.
 - Ova are removed from the ovary by laparoscopic surgery, a minimally invasive technique that only involves making a small incision.
 - The ova are placed in a glass container, sperm is added and they are incubated for about 24 hours for fertilisation to take place.
 - The fertilised eggs (zygotes) are transferred to a growth medium to ensure they can divide and grow into embryos.
 - Embryos that develop normally are identified. A small number (two to four) are placed in the woman's fallopian tube or can be implanted into the womb.
 - Embryos not used immediately may be frozen (in liquid nitrogen) for later use.
- 3 Give two non-medical options for an infertile couple.
 - 4 IVF is a time-consuming and expensive procedure that is not always successful, even after multiple attempts. Discuss why scarce medical resources are used on these procedures when some people have to wait a long time for treatment for life-threatening conditions.
 - 5 Do you think a child should be told that they were conceived using IVF? Explain your answer.
 - 6 Many embryos are created during the IVF process. Frequently there are some 'left over' when a procedure for a couple is successful. What should happen to these embryos?

Louise Brown was the first person to be born using IVF. Two British doctors, Patrick Steptoe and Robert Edwards, pioneered the process used. Louise also had a sister who was conceived using IVF. Louise has been the centre of a great deal of publicity since her birth in 1978. In 2003 she attended a reunion party at the clinic where IVF was first developed. Today the procedure is carried out in many clinics around the world. It can cost around \$20 000 to \$30 000 for a single IVF cycle with no certainty of conception.



- 7 Imagine you are Louise Brown. Someone says to you, "Hey, you're the girl who was born in a test tube!". Outline your response to this statement.
- 8 Discuss whether or not the cost of IVF treatment should be subsidised by government.



Surrogacy

Surrogate mothers are women who carry the implanted, fertilised zygote of another couple, usually because the biological mother cannot undergo the pregnancy. The surrogate mother commonly has her medical bills paid and receives a final payment after the child is born and given to its parents. Cases have occurred where the surrogate mother changes her mind and wishes to keep the baby rather than accept payment for carrying it (this includes refunding all medical expenses).

- 9 Do you agree or disagree with the concept of surrogacy? Give your reasons.
- 10 Give reasons why the surrogate mother might change her mind.
- 11 Which woman do you consider the 'real' mother?
- 12 If you were a judge, whose case would you support, the surrogate mother or the biological mother? Why?

Pre-implantation Genetic Diagnosis (PGD)

The term 'genetic screening' is becoming familiar to most people these days. PGD is a technique that works with standard IVF methods to check whether the embryos that are developing for implantation are carrying genes for genetic disorders.

- 13 Outline what is meant by 'IVF'.

Prior to implantation in the mother, embryos are tested, usually on day three of fertilisation. When a piece of tissue is removed from a living organism, to be examined under the microscope, the procedure is called a *biopsy*. To obtain the material needed to test the embryo, an *embryo biopsy* is performed. This is done by carefully removing a single cell from the embryo and it is believed that the procedure causes no ill-effect to its development. The embryo is returned to the petri dish to continue growing while tests are performed on the DNA within the extracted cell. This will take from several hours to a whole day but results are usually obtained within six hours. If no evidence is present for genetic disorders, the embryo will be one of those implanted in the mother.

- 14 If an embryo tests 'clean' does this mean a pregnancy will result? Explain your answer.
- 15 If the embryo tests positive for a disorder, the family is presented with the decision about it. Pro-life supporters believe that this is no less fair than any other abortion while others feel it is the kinder option from both the parents' and child's point of view. Which view do you support and why?
- 16 Outline the difference between an 'embryo' and a 'foetus'.

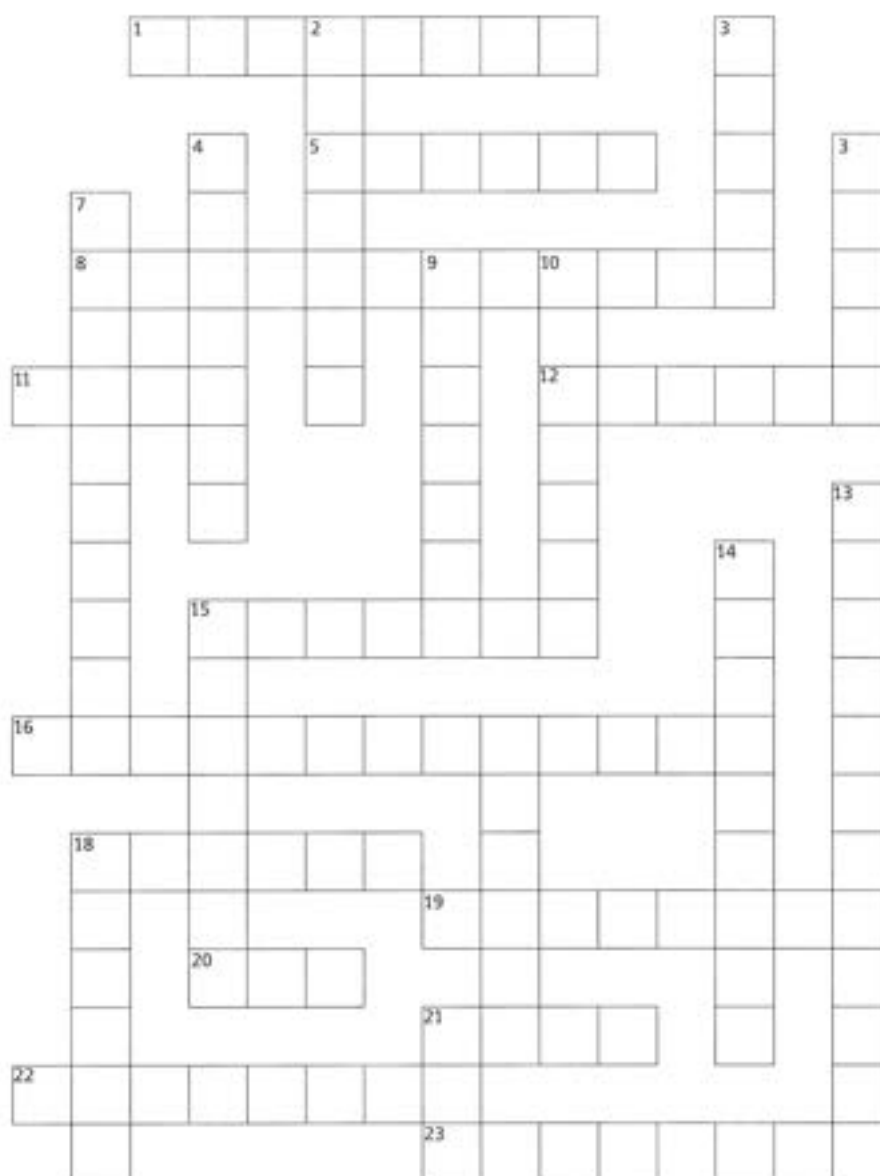
X-linked diseases

A number of X-linked diseases exist and many are associated with muscle disorders or *myopathy* (*myo* muscle, *pathos* disease). While not all of the muscular dystrophies are X-linked, many have at least one X-linked variation. Examples include:

- Duchenne Muscular Dystrophy (DMD)
- Emery-Dreifuss Muscular Dystrophy (EDMD)
- Spinal Bulbar Muscular Atrophy (SBMA)
- Charcot-Marie-Tooth Disease (CMT)
- Myotubular Myopathy (MTM)

- 17 Some of the conditions above are fatal while others, such as EDMD, are benign. Give a definition for *benign*. Compare it with *malignant*.
- 18 Describe the difference between a *sex-linked* disease and a *sexually transmitted* disease.
- 19 Write a paragraph describing one of the X-linked muscular disorders named above.

Blueprint of life crossword



Across

- 1 'Strong genes' which determine phenotype.
- 5 Produced when genetic material from two species is mixed.
- 8 Containing genes for two different characteristics.
- 11 When a baby is pushed out of the uterus into the world.
- 12 An organism which produces live young.
- 15 Cell division, producing cells with half the number of chromosomes.
- 16 This happens when a sperm and ovum fuse.
- 18 Preserved remains of an organism in rock.
- 19 Complete genetic makeup of an individual.
- 20 _____ chromosomes determine the gender of a baby.
- 21 Decomposers feed on _____ and decaying matter.
- 22 A sudden change in a chromosome or gene.
- 23 Reproduction without sexual processes.

Down

- 2 To pass on from one generation to another.
- 3 Formed from two fertilised eggs, or one egg which divides into two.
- 4 When a species no longer exists.
- 6 Brightly coloured part of flowering plants.
- 7 Twenty-three pairs of this contain genetic information (in humans).
- 9 Fertilised eggs.
- 10 Sex cells
- 13 The shape of a DNA strand.
- 14 Physical characteristics.
- 15 Cell division, producing two cells each with full set of chromosomes.
- 17 Alternative forms of a gene.
- 18 These organisms reproduce using spores.
- 21 Carries the genetic information in chromosomes.



1 Reproduction in organisms

- 1 unicellular
- 2 Minute, require microscope to see them.
- 3 Part of environment in which an organism lives.
- 4 They would gradually die out as resources were used up or evolve to adapt to other resources.
- 5 Contents of cell replicate and then separate from each other to produce clones.
- 6 2^{36}
- 7 Those figures are under optimal conditions, which are rare. Normally other outside factors would arise to reduce number of bacteria.
- 8 Moist, warm conditions.
- 9 Fern, fungi, mosses etc.
- 10 Progeny, children.
- 11 Wind, on animals' fur etc.
- 12 Bacteria, amoeba.
- 13 clone
- 14 one
- 15 two
- 16 Species is able to survive in varying habitats depending on its genes, stronger individuals pass on strong characteristics, etc.
- 17 Advantages: genetic diversity; disadvantages: slow, wasteful.
- 18 A fertilised cell.
- 19 Mature sex cells, sperm and egg.
- 20 Genetic material varies from parents: half each.
- 21 Many die before successful fertilisation.
- 22 eg human, dog, cat, elephant, dolphin etc.
- 23 eg chicken, eagle, crocodile, snake etc.
- 24 Young organism in early stages of development.
- 25 *vivi* live, *ovi* egg, *parous* bearing or producing
- 26 Slow, wasteful
- 27 Drought, climatic change etc.

2 Reproduction in plants

- 1 Part of stamen, contains pollen.
- 2 pollen
- 3 Produces seeds.
- 4 ovary
- 5 Part of pistil, receives pollen.
- 6 ova
- 7 Inner, modified leaves.
- 8 sepal
- 9 To attract insects.
- 10 To improve chance of wind carrying them away.
- 11 eg wind, people and animals brushing past, animals and birds eat fruit and pass seed.
- 12 Plants with seeds in a closed ovary eg flowering plants.
- 13 Surrounding conditions, other organisms and their influence on a given organism.
- 14 stamen
- 15 pistil
- 16 cross-pollination
- 17 Male and female parts reach maturity at different times; have male and female plants.
- 18 Much of the pollen doesn't reach ultimate destination.
- 19 Colour, scent, nectar, large petals.
- 20 eg good vision, fuzzy (so pollen sticks), pouches to store pollen.

- 21 Strongly scented, light colour – white is easily visible at night.
- 22 For natural pollination.
- 23 hay fever
- 24 Suitable bees or birds are not available; it is quicker and more accurate.

3 Human reproduction I

- 1 ova
- 2 Male gamete.
- 3 Female gamete.
- 4 Mature sex cell.
- 5 Internal mammalian organ to protect developing young.
- 6 Time taken from fertilisation to birth.
- 7 Developing young up to about seventh week (in humans).
- 8 Developing young from about third month to birth (in humans).
- 9 Union of male and female gamete.
- 10 pituitary
- 11 puberty
- 12 menstruation, 'period'
- 13 hormone
- 14 larynx
- 15 Broader chest, voice deepens, facial hair and pubic hair grows.
- 16 Hips broaden, breast development, pubic hair grows, menstruation begins.
- 17 Usually between the ages of 10 to 16 years.
- 18 true
- 19 false
- 20 Oxygen and nutrients are supplied through the placenta, waste products are also removed through the umbilical cord and placenta. Developing baby is cushioned in thickened wall of uterus and suspended in amniotic fluid.
- 21 a 80.5% b 18.1% c 1.25%

4 Human reproduction II: Birth

- 1 No, blood group is partially due to father's genes.
- 2 Numbs pain; reduces shock to the body.
- 3 belly button
- 4 hormones
- 5 umbilical
- 6 placenta
- 7 caesarean
- 8 cervix
- 9 amniotic
- 10 uterus
- 11 delivery
- 12 Avoid drugs, alcohol (including caffeine) and smoking; maintain healthy lifestyle – eat sensibly and exercise regularly etc; avoid stress and heavy lifting.
- 13 In order of appearance: cells, life, organism, microscope, animal, nucleus, cytoplasm, sperm, ovum, fertilisation, zygote, ovaries, oviduct, testes, penis, uterus, divide, implantation, embryo, grows, amniotic, placenta, oxygen, umbilical, navel, baby, cervix, afterbirth

5 The endocrine system

- 1 They enhance performance unfairly. 2 male
- 3 sperm and steroid hormones 4 pituitary 5 both

6 uterus 7 female 8 ova and steroid hormones
9 The organ which produces sperm (male) or eggs (female) in animals. 10 To be checked.

6 Methods of contraception

1–12, 14–2, 9–3, 8–4, 5–17, 6–13, 18–7, 10–16, 15–11

7 Human reproduction review quiz

1 T 2 T 3 T 4 T 5 T 6 F 7 T 8 T 9 F 10 F
11 F (organ) 12 F 13 F 14 T 15 F 16 F 17 T 18 F
19 T 20 F

8 Human senses

1 Touch, heat, light, warmth etc.
2 contracts
3 eg secretes a substance
4 eye 5 skin 6 food 7 nose 8 touch 9 ear 10 B
11 To deprive a sense of any stimulus, for example deprive vision of light and eyes will react to be light sensitive.
12 To be checked. 13 chemicals 14 tongue, nose
15 touch 16 skin 17 movement/pressure
18 skin 19 light 20 eyes 21 heat 22 skin
23 To make information known between two or more organisms.
24 Eg cry, whistle, wave, facial expressions etc
25 true 26 true 27 false 28 To be checked.

9 Vision I

1 Wavelength should start and finish at equivalent points.
2 tears
3 Animal that hunts other animals for food, eg lion, eagle, feral cat, crocodile.
4 eyes 5 brain 6 gland
7 Eyes at side give wider field of vision; eyes at front judge distances better.
8 Teacher to check.
9 1 cornea 2 pupil 3 iris 4 sclera 5 optic nerve
6 retina 7 vitreous humour 8 lens 9 muscles
10 blind spot
10 Light can't pass through; cloudy.
11 iris 12 retina 13 vitreous humour 14 optic nerve
15 sclera (sclerotic coat)
16 To magnify, aid myopia, hyperopia and cataracts.
17 lachrymal glands
18 Removes dust, lubricates.
19 In science experiments, using power tools.
20 Teacher to check, but should mention it is area of retina that is not sensitive to light.

10 Vision II

1 Colouring agent, dye.
2 Cones at centre of retina, rods at edges of retina.
3 cones
4 About 6 million.
5 Absorbs light/photosensitive pigment.
6 Red, blue and green.
7 Nocturnal – active at night; diurnal – active in daytime.
8 For better vision when hunting at night.
9 To recognise coloured fruit and to see when it is ripe.
10 Part of cell that contains most of genetic information.

11 Carried on a chromosome that is also responsible for assigning sex.

12 X^CY 13 X^CY 14 X^CX^C or X^CX^c 15 X^cX^c

16 Where C is (dominant) gene for normal sight and c is (recessive) gene for colour blindness.

17 Teacher to check, but should be consistent with conventions used in previous answers.

18 1 (rectus) muscle 2 sclera/sclerotic coat 3 iris 4 pupil

19 Transparent front of eyeball.

20 Protects and holds eye together.

21 Retina

22 Control amount of light entering the eye.

23 Lens

24 Watery liquid containing sugars, salts and proteins.

25 Keeps shape of eyeball, helps refract (bend) light.

26 Nerve cell

27 false 28 true 29 true 30 false 31 true

11 The human ear

1 In order of appearance: three, pinna, auditory, membrane, sound, ossicles, magnify, oval, eustachian, cochlea, neural, hairs, cells, high, basilar, nerve, brain, hear, hertz, ultrasonic
2 ear/auditory canal
3 middle ear
4 Equalises air pressure
5 'hammer'
6 middle ear
7 Connects to eardrum/magnifies movement.
8 oval window
9 inner ear
10 balance
11 auditory nerve
12 true 13 false 14 false 15 true 16 false
17 Sound reaches ears with different intensity and can be used to locate source of sound.

12 Sound and communication

1 twenty (20) cycles per second; ten thousand (10 000) cycles per second
2 tone
3 Low frequency: fewer waves and longer distances between peaks/troughs. High frequency: more waves and shorter distances between peaks/troughs.
4 larynx 5 trachea 6 cartilage 7 voice box
8 To warn others of danger, attract mates, give location.
9 decibel
10 bats, dolphins etc
11 whales
12 false 13 false 14 true 15 true
16 Teacher to check. Table should have clear headings and contain all examples.
17 Beyond range of normal hearing.
18 Sound waves are conducted by bones in skull.
19 faster

13 The senses – definitions

1 organism 2 stimulus 3 cortex 4 endocrine system
5 hormone 6 involuntary 7 neurone 8 ossicles
9 receptor 10 ear 11 taste buds 12 skin 13 retina

14 focus 15 brain 16 cochlea 17 adrenaline
18 pituitary 19 electrical 20 autonomic 21 pupil
22 optic 23 binocular 24 effector 25 cornea
26 lens 27 electromagnetic spectrum

14 The senses review quiz

1 D 2 A 3 B 4 A 5 B 6 A 7 D 8 B 9 D 10 C 11 C
12 C 13 C 14 B 15 true 16 true 17 true 18 true 19 false
20 true 21 true 22 true 23 true 24 false 25 false
26 true 27 true 28 false 29 cones 30 retina
31 neurones 32 central nervous system
33 voice box/larynx 34 brain 35 myopia (short sighted)
36 refraction 37 sight 38 accommodation
39 colour blindness 40 eustachian

15 Historic figures in genetics

- 1 A particular characteristic.
- 2 To make sure he knew exactly where the pollen of each plant went and could accurately record results.
- 3 To collect and classify facts based on numbers.
- 4 Identical individuals that produce offspring resembling themselves.
- 5 Anatomy/Morphology
- 6 Taxonomy
- 7 Recessive traits can be hidden; other factors may change physical features eg sun bleaches hair, artificial dyes etc.
- 8 Haploid sex cell.
- 9 Physical appearance.
- 10-12 To be checked.
- 13 a Alleles are bands labelled 'A' and 'a'.
b Centromere is circle.
c Chromatid is either one of the two strands.
- 14 centromere
- 15 false 16 true 17 true 18 false 19 false 20 true
21 false 22 false 23 true 24 true

16 Review of genetics terms

gene: basic unit of inheritance
allele: alternative form of a gene
dominant: overriding allele
recessive: masked in phenotype
homozygous: same
heterozygous: different
genotype: complete genetic make up of an organism
phenotype: physical appearance
lethal: kills
dominant lethal: lethal in a single dose
conditional lethal: lethal under certain conditions only
chromosomes: material carrying genes
chromatids: replicated chromosomes
centromere: site at which sister chromatids join
autosome: non-sex chromosome
sex (X) chromosome: chromosomes associated with sex determination
autosomal lethal: lethal found on non-sex chromosome
sex-linked lethal: lethal found on sex chromosome
X-linked recessive: expression (or not) determined by organism's sex
X-linked: linked to sex chromosome

17 Punnett squares, peas and crossing

- 1 Identical alleles for a trait. 2 Two different alleles.

- 3 RR, Rr, Rr, rr 4 RR – round, Rr – round, rr – wrinkled
- 5 Round to wrinkled = 3:1
- 6 Second filial (F₂)
- 7 Reading across each row from left to right:
Row 1: all will be round/green
Row 2: round/green, round/yellow, round/green, round/yellow
Row 3: round/green, round/green, wrinkled/green, wrinkled/green
Row 4: round/green, round/yellow, wrinkled/green, wrinkled/yellow

18 Role and structure of chromosomes

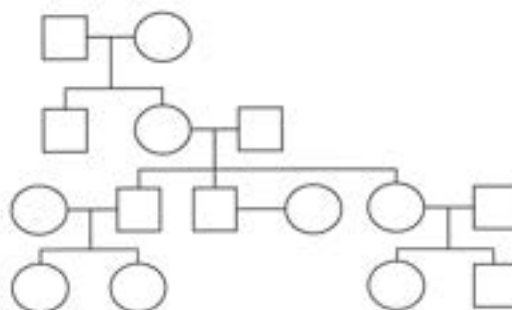
- 1 two
- 2 More developed countries have the money and therefore the technology for GM crops.
- 3 Basic structural units of nucleic acids.
- 4 Hydrogen bonding between opposite groups on each strand.
- 5 Destroy properties.
- 6 DNA
- 7 genetic engineering
- 8 cloning
- 9 chromosome

19 Genetic issues in the 21st century

- 1 Drugs used for medicinal purposes.
- 2 Can change location.
- 3 Taken by mouth.
- 4 Worried that not enough research has been done; products are not natural and therefore may be harmful.
- 5 They can be beneficial to humans.
- 6 Ethics/morals involved in genetic research.
- 7 Study of genetic make up of an organism.
- 8 Study of proteins of an organism.
- 9 Interbreed, produce hybrids.
- 10 Breeding to get desired characteristics.
- 11 Breeding to create hybrids.
- 12 Separate, remove.
- 13 copied 14 joining/joint 15 female sheep 16 treatment
- 17 Eg cattle or sheep bred to produce more milk/meat/wool etc.
- 18 Cattle/sheep of different breeds to get mix of desired characteristics, eg creamier milk, softer wool.
- 19 Teacher to check. Traditional uses pesticides and artificial fertilisers etc.
- 20 To be checked. Should resemble illo on page 32.

20 Pedigrees

- 1 See diagram.



- 2 Direct – born into a family; collateral – married into a family.
- 3 Punnett squares
- 4 First, second and third generations.
- 5 Male without phenotype marries a female with phenotype. They have three children, two males without phenotype and a female with phenotype. One of the males marries a female without phenotype and they have three children – a male without phenotype, a male with phenotype and a female without phenotype.
- 6 Only one dominant gene needs to be present for associated phenotype to appear, whereas two recessive genes need to be present for phenotype to appear.
- 7 They have two X chromosomes.
- 8 Having two different genes for a characteristic.
- 9 gonad
- 10 hermaphrodite
- 11 parthenogenesis
- 12 Male – XY chromosomes, produce sperm, etc; female – XX chromosome, produce ova. Could also mention secondary characteristics such as facial hair, voice etc.
- 13 Teacher to ensure that a range of answers are given and that there is an appreciation that roles of men and women vary between societies.
- 14 vestigial: diminished, unviable: not viable
- 15 Most have a male phenotype, look more male.
- 16 Attracted to the same sex though still identify with own sex.
- 17 Tend to be male, dress as female and appear undecided as to their true sex.
- 18 Physically one sex but mentally feel like the other, may dress accordingly.

21 Genetics – definitions

- 1 evolution 2 resources 3 embryology
- 4 characteristic 5 inheritance 6 heterozygous
- 7 alleles 8 dominant 9 pedigree 10 chromosomes
- 11 diploid 12 meiosis 13 gamete 14 fertilisation
- 15 co-dominance 16 nucleotide 17 polypeptide
- 18 mutation 19 genetic engineering 20 cloning
- 21 transgenic 22 Sutton 23 double helix
- 24 Punnett 25 replication 26 hybrid

22 Medical ethics

(Note there are no 'right' answers in this section. Answers given are samples only.)

- 1 Western – privacy, no outside interference; Eastern – group decision, input from family etc.
- 2 Teacher to check – logical answer expected.
- 3 Whether to allow pregnancy to continue; whether to have further babies etc.
- 4 Availability of medical treatment/care, cost, effect on existing siblings.
- 5-6 Teacher to check – logical argument expected.
- 7 Teacher to check, eg if the mother is in an irreversible coma.
- 8 Society has an obligation to care for disadvantaged members, if they (or their family) cannot fund treatment.
- 9 For: great medical breakthrough, saved many lives. Against: risking life of a person.

23 In vitro fertilisation

- 1 Inherited causes; low sperm count/sperm motility; illness.
- 2 Ova not viable; production of ova not regular.
- 3 Remain without children, adoption, surrogacy.
- 4 Eg desire to have children is very strong.
- 5 Teacher to check – eg child has 'right to know' vs effects of psychological shock.
- 6 Teacher to check – eg use in other procedures or for medical research vs destroy.
- 7 Eg "No, an egg may have been fertilised in a test tube, but I was born in a clinic, in the normal manner, from my mother" etc.
- 8 Yes – too expensive for individuals, people's right to have children. No – more urgent medical problems need funding.
- 9 Teacher to check – eg natural mother needs to pay bills etc vs shouldn't be able to 'buy a baby'.
- 10 Eg form attachment to baby, have moral dilemma etc.
- 11-12 Teacher to check – logical argument expected.
- 13 In vitro fertilisation – (literally 'in glass') fertilisation of an egg with sperm outside of the body.
- 14 No, there are many other considerations.
- 15 Teacher to check – logical argument expected.
- 16 Embryo is less developed (up to 8 weeks from fertilisation in humans) than foetus.
- 17 Benign doesn't threaten health, malignant does.
- 18 Sex-linked is something that is inherited on either X or Y chromosomes. Sexually transmitted means a condition passed on through sexual intercourse.
- 19 Teacher to check – should include symptoms.

24 Blueprint of life crossword



Instant Lessons

Quality Ready-to-Use Resources

Each book in the **Instant Lessons** series is a collection of fresh and versatile worksheet activities which may be photocopied for student use. They are unique in providing the **highest quality learning outcomes with instant usability**. Below is a checklist of the major uses and benefits of these books.

Versatile Uses		Teaching Benefits	
Supplementary classroom lessons	✓	Creative teaching ideas	✓
Homework activities	✓	Developing greater student interest and understanding	✓
Reinforcement of learning	✓	A fresh approach to a topic or lesson	✓
Extension work for advanced students	✓	A way of targeting student weaknesses	✓
Preparation for assessment	✓	Suitable for individual or group work	✓
Exam revision	✓	A variety of activities for home study	✓

Use them to supplement your existing resources, to extend student capabilities, for revision purposes, or for class work when the regular teacher is absent.

Assignments in Biology Book 2

This book aims to provide teachers with innovative lessons across a variety of topics studied in Biology at the senior secondary level. Book 2 covers Reproduction, the Senses, Genetics and Medical Ethics. Through a range of activities, scientific knowledge is tested, and literacy and research skills are also developed. Answers have been provided to make marking of student work easy and time-efficient.

Topics include:	
Reproduction in organisms Human reproduction Methods of contraception Human senses Vision The human ear Sound and communication	Historic figures in Genetics Genetic terms Punnett squares, peas and crossing Role and structure of chromosomes 21st century issues Pedigrees Medical ethics

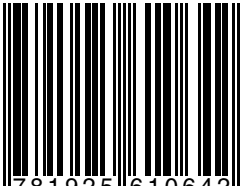
Associated titles in the Instant Lessons series

- Instant Lessons in Biology (Books 1-3)
- Assignments in Genetics (Books 1 & 2)
- Interpreting Data in Biology (Books 1 & 2)
- Assignments in Physics (Books 1 & 2)

Copyright notice

Permission is given to schools or teachers who buy this book to make photocopies of all pages. The copies must be for internal school use only, and may not be given or sold to other educational institutions or to teachers from other institutions.

ISBN: 978-1-92561-064-2



Instant Lessons
An imprint of Learning Essentials
A.B.N. 56 093 980 726
108 Main Rd
Clayton South VIC 3169
Ph: (03) 9509 5366
www.learningessentials.com.au

learning
ESSENTIALS
Ready-to-Use Resources